

CLIMATE CHANGE VULNERABILITY AND RISK ASSESSMENT REPORT FOR BUNAMWAYA DIVISION 2025 - 2030

JUNE 2025

Approved by the Council under
Minute Number MSMC/COU/03/26/06/25

EXECUTIVE SUMMARY

Climate change poses a growing threat to both natural ecosystems and human settlements in Uganda, with urban and peri-urban areas like Bunamwaya Division in Makindye Ssabagabo Municipality experiencing increasing climate-related vulnerabilities. Characterized by rapid urbanization, informal settlements, degraded wetlands, and poor drainage infrastructure, Bunamwaya is particularly exposed to the adverse impacts of climate hazards such as flooding, prolonged dry spells, and increased temperatures. This Local Climate Vulnerability and Risk Assessment (CVRA) report aimed to identify and analyze the climate-related risks affecting the communities, infrastructure, and ecosystems in Bunamwaya Division.

The methods that were used to develop the CVRA for Bunamwaya division included literature review, stakeholder engagements, multi-hazard mapping and profiling, exposure, vulnerability and risk assessment. The key stakeholders that were consulted and validated this report included the assistant town clerk, town agents, and community development officer.

Climate change in Bunamwaya division has manifested through increased temperatures, increased extreme events such as floods, prolonged dry spells, unpredictable and erratic rainfall patterns, increased amount of rainfall, and change of planting seasons. The climate hazards that occur in Bunamwaya division include drought, floods, lightning, hailstorms, and windstorms. These hazards have impacted the agriculture, education, energy, health, land and housing, manufacturing, social development, trade, transportation, and water and environment sectors of Bunamwaya division.

The areas that have a high level of exposure (more than 80 percent) to all the climate hazards in Bunamwaya division include Mirembe, Wabiyinja, Kanaala, Busingiri Nyanama, Nakinga, Lufuka, Kirimanya, and Zana. The cells that are highly vulnerable to all the hazards include Mirembe, Nakinga, Lufuka, Kirimanya, and Zana. Similarly, the cells at risk of all hazards in Bunamwaya division include include Mirembe, Wabiyinja, Kanaala, Kisigula, Nakinga, Busingiri Nyanama, Lufuka, and Kirimanya.

Therefore, the government with support from partners should design innovative approaches to reduce the vulnerability and risk of these communities to multi-hazards and also increase their resilience and livelihoods. Some of those approaches may include;

- develop and maintain a local climate risk database to support evidence-based planning
- integrate hazard-specific data into local development plans
- upgrade and regularly maintain flood-prone drainage systems and culverts
- Promote stronger building materials, elevated housing, and storm-resistant roofing

TABLE OF CONTENTS

1. EXECUTIVE SUMMARY i

2. LIST OF FIGURES iv

3. LIST OF TABLES v

4. LIST OF PLATES v

1. INTRODUCTION 1

1.1 Background 1

1.2 Purpose and Objectives 1

1.2.1 Purpose 1

1.2.2 Objectives 1

2. DESCRIPTION OF BUNAMWAYA DIVISION 2

2.1 Location of Bunamwaya division 2

2.2 Socio-demographic characteristics 3

2.3 Wetland Ecosystems 4

3. CLIMATE CHANGE IN BUNAMWAYA DIVISION 6

3.1 Indicators of climate change 6

3.2 Climate Trends and Projections 6

3.2.1 Observed and Projected Maximum Temperature Trends in Bunamwaya division 6

3.2.2 Observed and Projected Minimum Temperature Trends in Bunamwaya division 7

3.2.3 Observed and Projected Rainfall Trends in Bunamwaya division 9

3.3 Impacts of climate change in Bunamwaya division 10

4. VULNERABILITY AND RISK ASSESSMENT IN BUNAMWAYA DIVISION .. 11

4.1 Exposure to climate hazards 11

4.1.1 Drought Hazard 11

4.1.2 Flood Hazard 12

4.1.3 Hailstorm Hazard 14

4.1.4 Lightning Hazard 15

4.1.5 Windstorm Hazard 16

4.1.6 Multi-hazard 18

4.2 Sensitivity of communities to multi-hazards 19

4.3 Adaptive capacity of communities to multi-hazards 22

4.4 Vulnerability of communities to multi-hazards 26

4.5 Risk of communities to multi-hazards 26

5. RISK REDUCTION MEASURES TO CLIMATE HAZARDS 28

5.1 Introduction 28

5.2 Hazard-specific Disaster Risk Reduction Measures 28

6. CONCLUSION AND RECOMMENDATIONS 30

6.1 Conclusion 30

6.2 Recommendations 30

7. REFERENCES 32

LIST OF FIGURES

Figure 1: Location of Bunamwaya division in Makindye Ssabagabo Municipality 2
Figure 2: Projected population in Bunamwaya division between 2015 and 2025 3
Figure 3: Sources of livelihood for communities in Bunamwaya division in 2014 3
Figure 4: Status of wetlands in Bunamwaya division between 2005 and 2025 5
Figure 5: Indicators of climate change in Bunamwaya division 6
Figure 6: Observed and projected maximum temperatures for Bunamwaya division from 1994 to 2030 7
Figure 7: Projected maximum temperature for the 2025-2030 near-term projection period in Bunamwaya division 7
Figure 8: Observed and projected minimum temperatures for Bunamwaya division from 1994 to 2030 8
Figure 9: Projected minimum temperature for the 2025-2030 near-term projection period in Bunamwaya division 8
Figure 10: Observed and projected rainfall for Bunamwaya division from 1994 to 2030 9
Figure 11: Projected rainfall for the 2025-2030 near-term projection period in Bunamwaya division 9
Figure 12: Drought hazard zonation in Bunamwaya division 12
Figure 13: Flood hazard zonation in Bunamwaya division 13
Figure 14: Hailstorm hazard zonation in Bunamwaya division 14
Figure 15: Lightning hazard zonation in Bunamwaya division 16
Figure 16: Windstorm hazard zonation in Bunamwaya division 18
Figure 17: Multi-hazard exposure to drought, floods, lightning, hailstorms, and windstorms in Bunamwaya division 19
Figure 18: Sensitivity of communities to multi-hazards in Bunamwaya division 22
Figure 19: Adaptive capacity of communities to multi-hazards in Bunamwaya division 25
Figure 20: Vulnerability of communities to multi-hazards in Bunamwaya division 26
Figure 21:The risk matrix 27
Figure 22: Risk of communities to multi-hazards in Bunamwaya division 27

LIST OF TABLES

Table 1: Impacts of climate change by sector in Bunamwaya division 10
Table 2: Lightning hazard intensity scale 15
Table 3: Beaufort wind hazard intensity scale 17
Table 4: Sensitivity indicators to multi-hazards in Bunamwaya division 20
Table 5: Adaptive capacity indicators to multi-hazards in Bunamwaya division 22
Table 6: Structural and non-structural DRR measures in Bunamwaya division 28

LIST OF PLATES

Plate 1: An intersection of solid waste disposal and infrastructural development in Lufuka wetland, Mutundwe parish, Bunamwaya division 4

1. INTRODUCTION

1.1 Background

Climate change poses a growing threat to both natural ecosystems and human settlements in Uganda, with urban and peri-urban areas like Bunamwaya Division in Makindye Ssabagabo Municipality experiencing increasing climate-related vulnerabilities. Characterized by rapid urbanization, informal settlements, degraded wetlands, and poor drainage infrastructure, Bunamwaya is particularly exposed to the adverse impacts of climate hazards such as flooding, prolonged dry spells, and increased temperatures.

This Local Climate Vulnerability and Risk Assessment (CVRA) report aimed to identify and analyze the climate-related risks affecting the communities, infrastructure, and ecosystems in Bunamwaya Division. The assessment provides an evidence-based foundation to inform climate adaptation planning, risk mitigation strategies, and policy integration at the division and local levels. The report applies a participatory and multi-disciplinary approach, integrating both scientific data and community-based knowledge to evaluate the exposure, sensitivity, and adaptive capacity of communities and ecosystems. Special attention is given to vulnerable groups such as women, youth, the elderly, and persons with disabilities, who often face disproportionate climate risks.

By focusing on localized impacts and risks, the CVRA supports Makindye Ssabagabo Municipality and key stakeholders in prioritizing adaptation actions, strengthening institutional coordination, mainstreaming climate resilience into development planning, and enhancing community awareness and preparedness. This assessment contributes to Uganda's broader national commitments under the National Climate Change Policy, the National Development Plan IV, and the Paris Agreement, while aligning with the local realities and urgent needs of the people of Bunamwaya.

1.2 Purpose and Objectives

1.2.1 Purpose

The report aims to identify, analyze, and document the key climate-related risks and vulnerabilities affecting the communities, infrastructure, and ecosystems of Bunamwaya Division, Makindye Ssabagabo Municipality.

1.2.2 Objectives

The objectives of the local climate vulnerability and risk assessment include;

I. To assess the current and projected climate hazards affecting Bunamwaya Division.
II. To identify the most vulnerable communities, groups, and assets in relation to climate risks.
III. To recommend priority adaptation actions, policies, and investments needed to reduce climate risks and enhance community and institutional resilience.

2. DESCRIPTION OF BUNAMWAYA DIVISION

2.1 Location of Bunamwaya division

Bunamwaya division is located in the Northern part of Makindye Ssabagabo Municipality. The division is bordered by Kampala district to the North and North East; Masajja division to the East; Ndejje division to the South and South East; and Wakiso district to the West (**Figure 1**). Bunamwaya division covers a total area of about 12.5 square kilometers. Administratively, the division has 2 wards (Bunamwaya, and Mutundwe) and 15 cells.

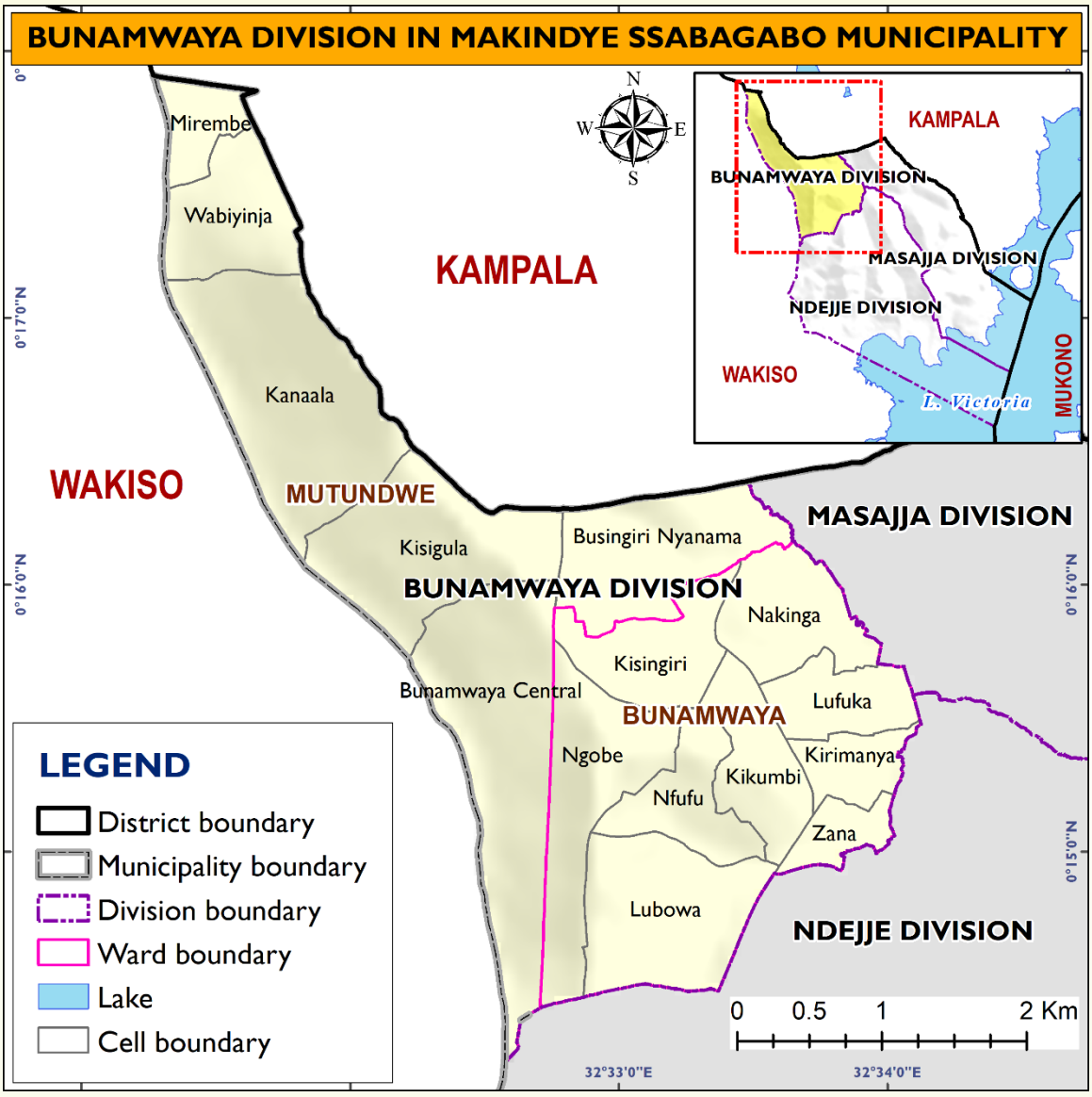


Figure 1: Location of Bunamwaya division in Makindye Ssabagabo Municipality

2.2 Socio-demographic characteristics

Bunamwaya division has a projected population of about 144,400 people (UBOS, 2025). Of these, 66,700 people (46.2%) are males whereas 77,700 people (53.8%) are females (**Figure 2**). Between 2015 and 2025, the population in Bunamwaya division is projected to have increased by 67,900 people (88.7%). In a rapidly urbanizing area like Bunamwaya, population growth contributes to climate change by driving wetland encroachment, increased energy use, and waste generation; all of which elevate greenhouse gas emissions and strain local ecosystems. The division has a population density of about 11,521 people per square kilometer, the highest in the municipality. The cells with the biggest population density include Kirimanyaga, Lubowa, Lufuka, Zzana, Busingiri-Nyanama, Wabiyinja, Kikumbi, and Kanaala.

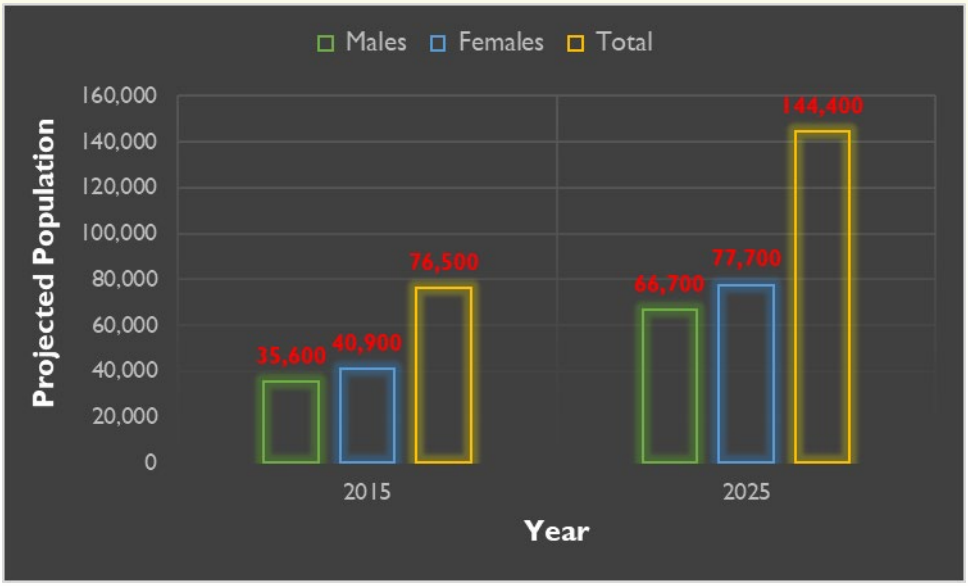


Figure 2: Projected population in Bunamwaya division between 2015 and 2025

The major sources of livelihood for communities in Bunamwaya division include business enterprises (58.4%), formal employment (11.9%), remittances (10.9%), casual labouring (7.9%), and transport services e.g., boda boda riding (4%) (**Figure 3**).

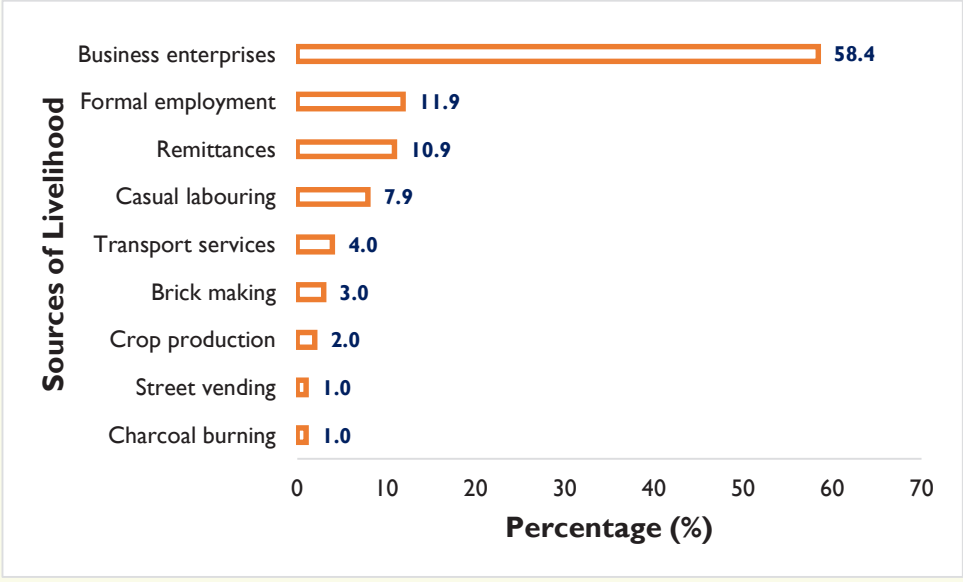


Figure 3: Sources of livelihood for communities in Bunamwaya division

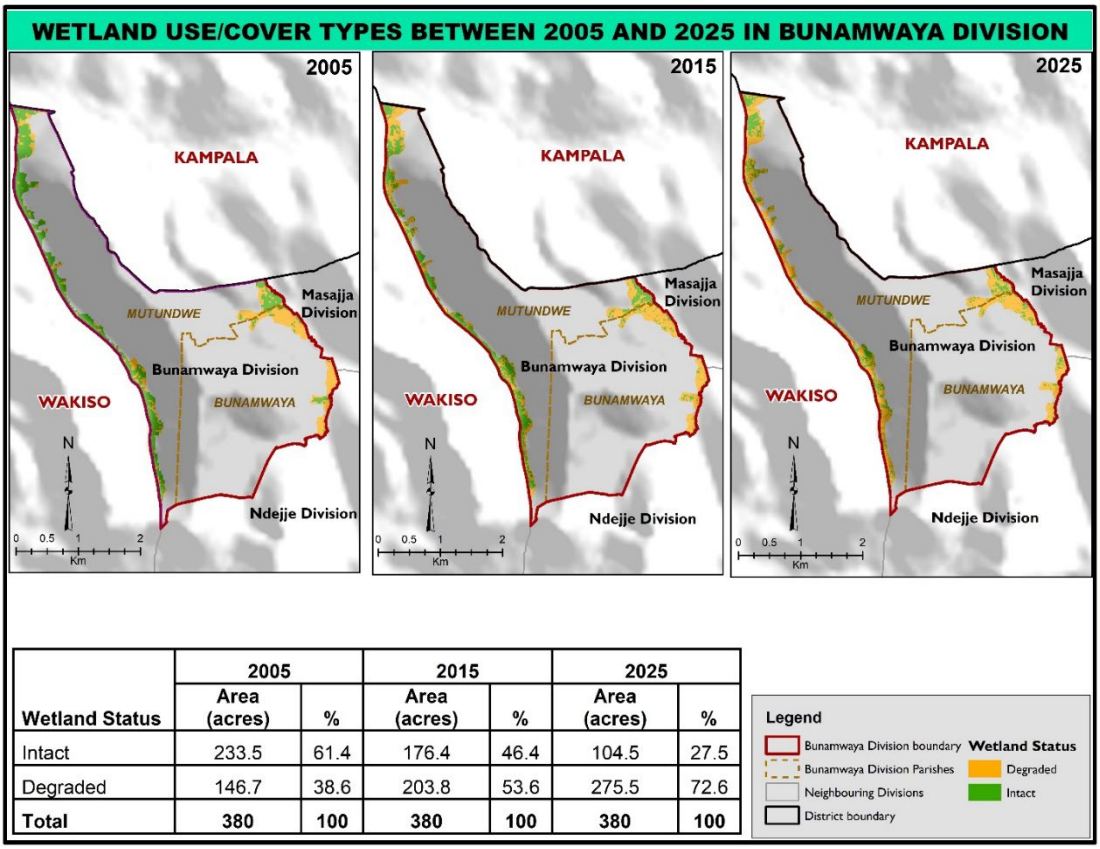
2.3 Wetland Ecosystems

Bunamwaya division has a total of 380 acres of wetlands (18% of the division surface area). The assessment showed that 72.5% of these wetlands (275.5 acres) are degraded whereas 27.5% of the wetlands (104.4 acres) are intact (**Figure 4**). The percentage of degraded wetlands in Bunamwaya division increased from 38.6% in 2005 to 72.5% in 2025. The cells with the biggest percentages of degraded wetlands include Kirimanya (90.9%), Kisingiri (85.8%), Zana (79.4%), Nakinga (77.3%), Kanaala (77.2%), Lufuka (72.2%), Wabiyinja (71.7%), and Busingiri Nyanama (71.4%). The major drivers of wetland degradation in the division include infrastructural development, dumping of solid waste in wetlands, brick making, sand and clay mining, and sand and clay mining.



Plate 1: An intersection of solid waste disposal and infrastructural development in Lufuka wetland, Mutundwe parish, Bunamwaya division

Figure 4 shows the status of wetlands in Bunamwaya division between 2005 and 2025.



Wetland Status	2005 Area (acres)	2005 %	2015 Area (acres)	2015 %	2025 Area (acres)	2025 %
Intact	233.5	61.4	176.4	46.4	104.5	27.5
Degraded	146.7	38.6	203.8	53.6	275.5	72.6
Total	**380**	**100**	**380**	**100**	**380**	**100**

Figure 4: Status of wetlands in Bunamwaya division between 2005 and 2025

Wetland degradation contributes to climate change by releasing large amounts of stored carbon into the atmosphere, reducing the land's natural ability to absorb greenhouse gases, and disrupting local water and temperature regulation systems. When wetlands are drained, filled, or polluted; they emit carbon dioxide and methane, potent greenhouse gases, and lose their function as natural carbon sinks, flood buffers, and temperature stabilizers.

3. CLIMATE CHANGE IN BUNAMWAYA DIVISION

3.1 Indicators of climate change

The communities were asked whether they have experienced climate change or variability in their areas. About 89.5% of the respondents reported experiencing climate change or variability in their respective wards. The indicators of climate change or variability reported by community members include increased temperatures (15.3%), increased extreme events such as floods and drought (12.4%), prolonged dry spells (11.2%), unpredictable and erratic rainfall patterns (10.2%), increased amount of rainfall (8.6%), change of planting seasons (7.1%), and increased occurrence of strong winds (6.8%) (**Figure 5**). The key informants also revealed that climate change in Bunamwaya has manifested through increased prolonged dry spells, increased temperatures characterized by too much heat, and change of planting seasons.

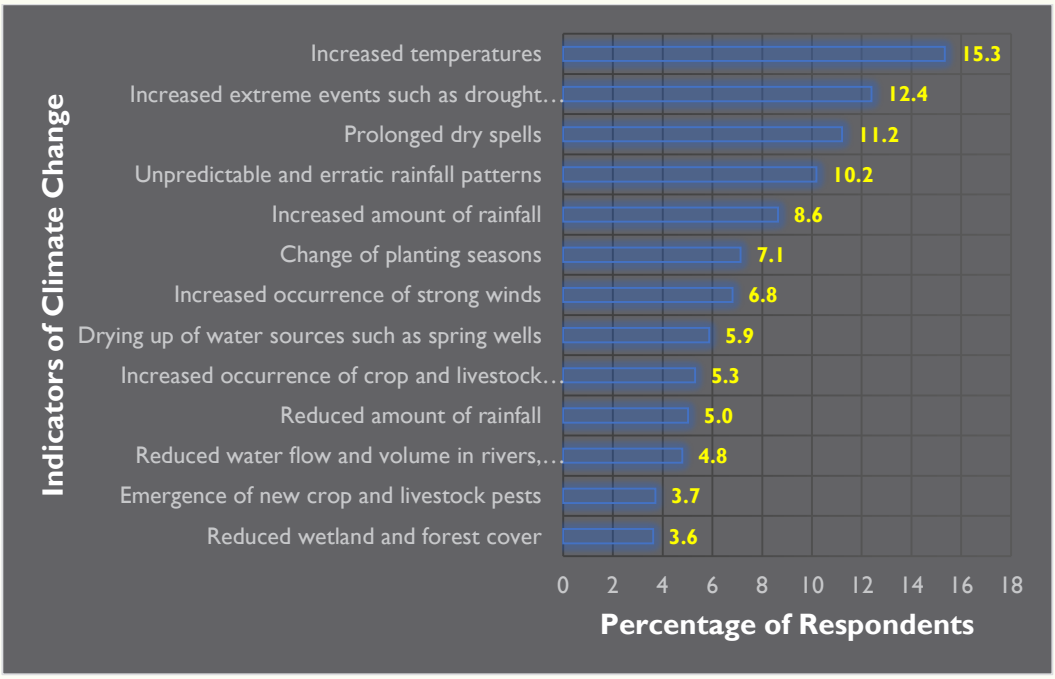


Figure 5: Indicators of climate change in Bunamwaya division

3.2 Climate Trends and Projections

3.2.1 Observed and Projected Maximum Temperature Trends in Bunamwaya division

The observed average annual maximum temperatures for Bunamwaya division are about 24.6 °C, with a minimum of 24.1 °C in August and a maximum of 25.9 °C in February. For the climatological period 1994–2024, maximum temperature in Bunamwaya division has increased at a rate of 0.23 °C per decade. Between 1994 and 2024, the highest maximum temperature of 25.3 °C was recorded in 2024 (**Figure 6**). The projections of maximum temperature (2025–2030) indicate an increase of 0.41°C and 0.64 °C under moderate (RCP 4.5) and high (RCP 8.5) greenhouse gas (GHG) concentration scenarios, respectively; relative to the 1994–2024 average.

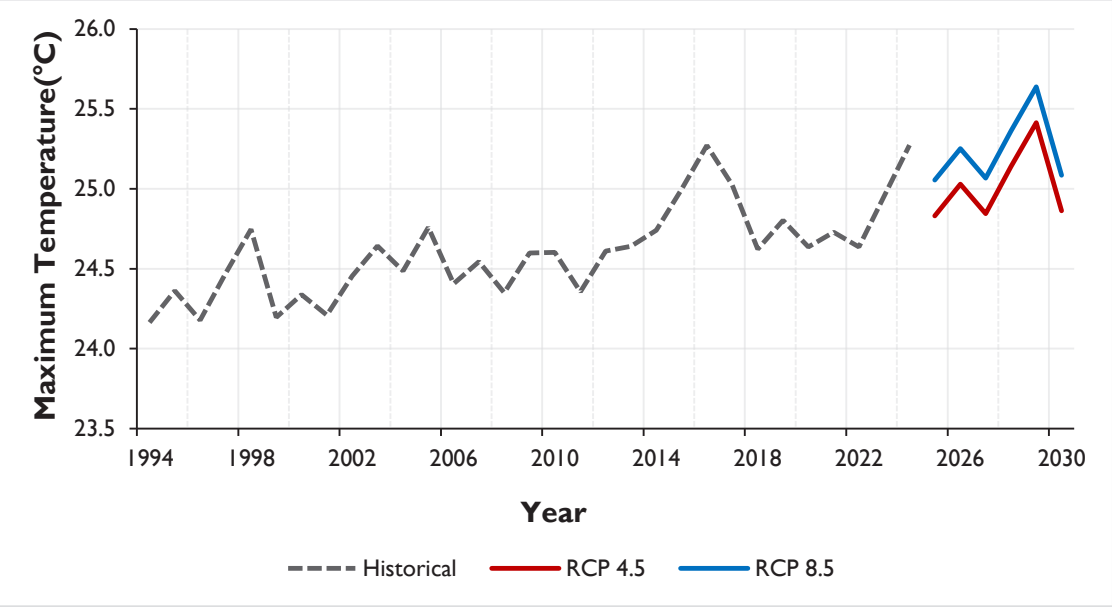


Figure 6: Observed and projected maximum temperatures for Bunamwaya division from 1994 to 2030

The highest projected maximum temperatures between 2025 and 2030 are expected to be experienced in Mirembe, Wabiyinja, Kanaala, Kisigula, and Busingiri Nyanama cells; especially under the high (RCP 8.5) greenhouse gas (GHG) concentration scenario (**Figure 7**).

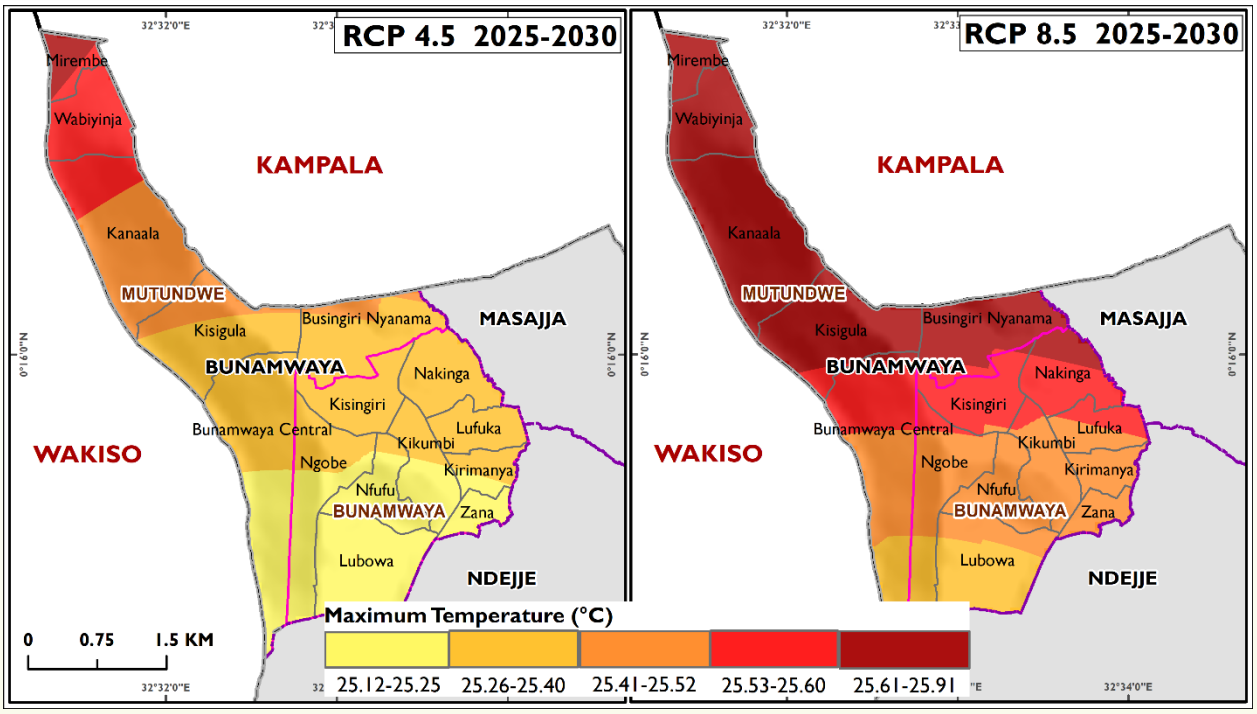


Figure 7: Projected maximum temperature for the 2025-2030 near-term projection period in Bunamwaya division

3.2.2 Observed and Projected Minimum Temperature Trends in Bunamwaya division

The observed average annual minimum temperatures for Bunamwaya division are about 20.8 °C, with a minimum of 19.6 °C in August and a maximum of 21.2 °C in February. Over the

climatological period 1994-2024, the minimum temperature has increased at a rate of 0.22 °C per decade. The highest minimum temperature of 20.9 °C was recorded in 2024 (**Figure 8**). Projections for the period 2025-2030 show an expected increase in minimum temperature of about 0.7 °C under RCP 4.5 and 0.6 °C under RCP 8.5, relative to the 1994-2024 average.

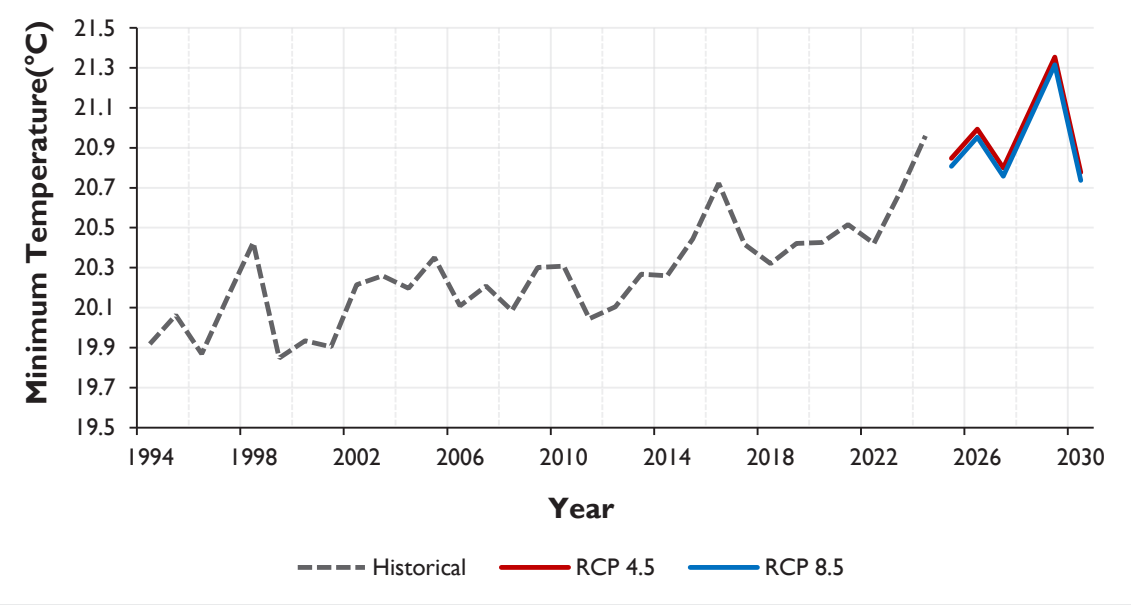


Figure 8: Observed and projected minimum temperatures for Bunamwaya division from 1994 to 2030

The areas of Zana, Lubowa, Nfufu, Kirimanya, Ngobe, Kikumbi and Lufuka are projected to have the highest minimum temperatures especially under the moderate (RCP 4.5) greenhouse gas (GHG) concentration scenario (**Figure 9**).

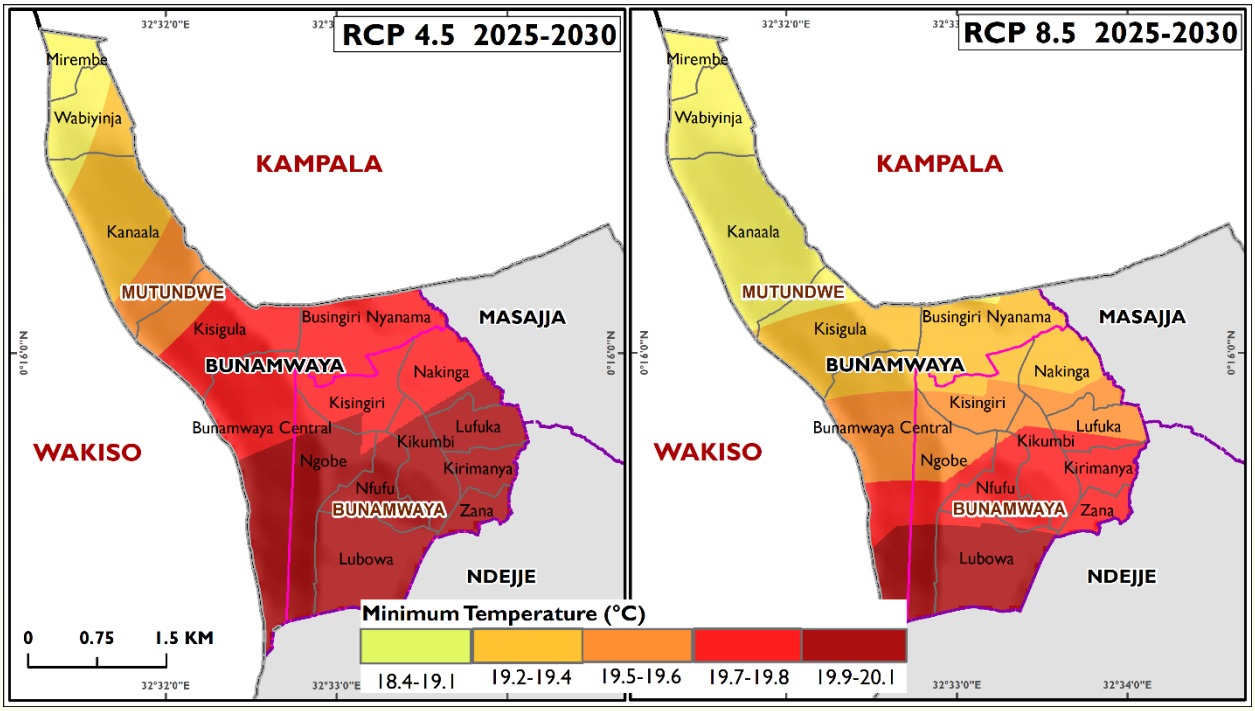


Figure 9: Projected minimum temperature for the 2025-2030 near-term projection period in Bunamwaya division

3.2.3 Observed and Projected Rainfall Trends in Bunamwaya division

Between 1994 and 2024, the total annual average rainfall for Bunamwaya division was 1,411.4 mm, and mean monthly rainfall varied from 59.8 mm in February to 218.9 mm in April. For the climatological period 1994-2024, rainfall in Bunamwaya division has decreased at a rate of 24.2 mm per decade. Between 1994 and 2024, the highest annual rainfall of 1,756.4 mm was recorded in 1997 (**Figure 10**). The projections (2025-2030) indicate an increase of mean rainfall for Bunamwaya division. Under RCP 4.5, mean rainfall is projected to increase by 271.1 mm (19.2%) relative to the 1994-2024 average. Similarly, under RCP 8.5, mean rainfall is projected to increase by 245.6 mm (17.4%) relative to the 1994-2024 average. These projections suggest a potential shift toward wetter conditions in the near future under both climate change scenarios.

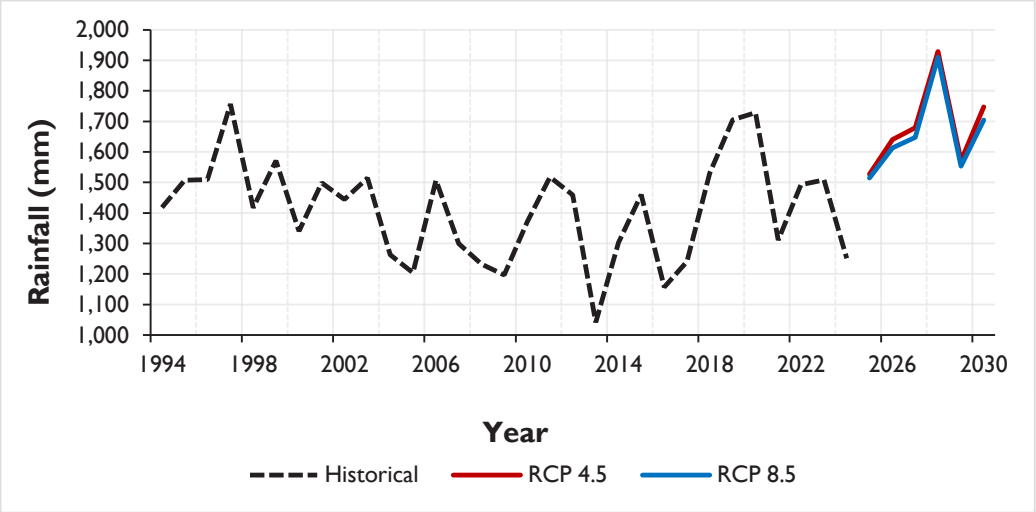


Figure 10: Observed and projected rainfall for Bunamwaya division from 1994 to 2030

The highest projected mean rainfall in Bunamwaya division is expected around Zana, Lubowa, Nfufu, Kirimanya, Ngobe, Kikumbi Bunamwaya Central, Kisingiri, Nakinga, and Lufuka cells (**Figure 11**).

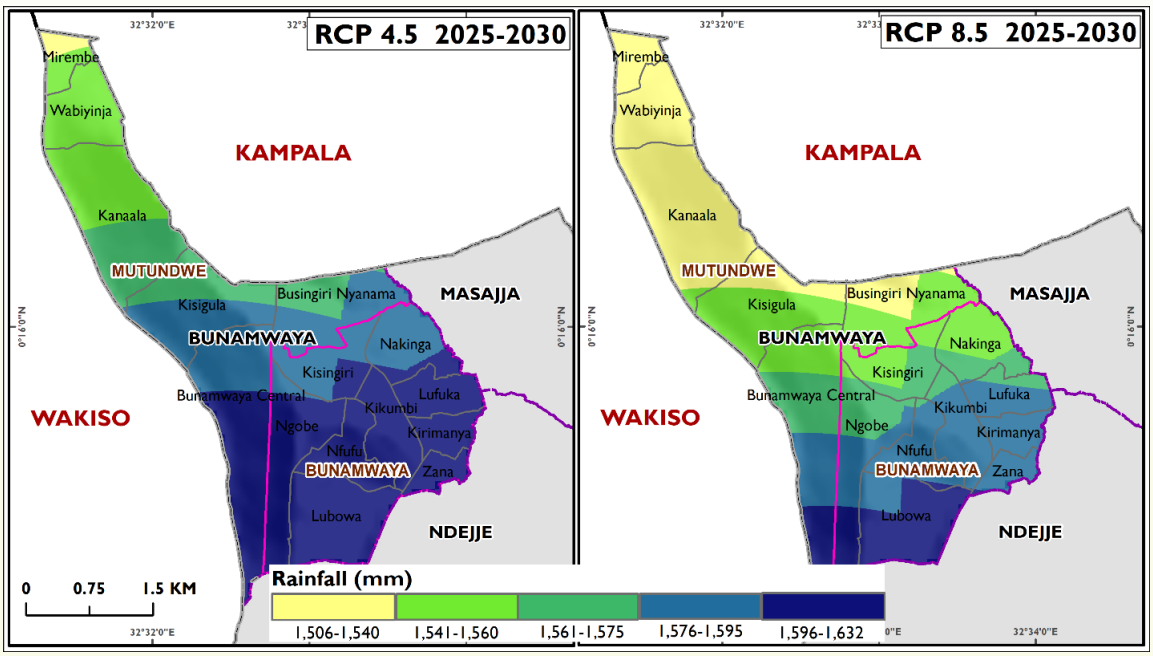


Figure 11: Projected rainfall for the 2025-2030 near-term projection period in Bunamwaya division

3.3 Impacts of climate change in Bunamwaya division

Climate change has significantly impacted the agriculture, education, energy, health, land and housing, manufacturing, social development, trade, transportation, and water and environment sectors of Bunamwaya division (**Table 1**).

Table 1: Impacts of climate change by sector in Bunamwaya division

Sectors	Impacts of climate change by sector
Agriculture	• Reduced crop yield • Reduced livestock productivity • Destruction of croplands • Increased prevalence of pests, parasites, and diseases • Reduced quality of forage available to grazing livestock
Education	• Damage to school buildings, rendering them unsafe or unusable • Displacement disrupts children's education and causes psychosocial stress • Disruption of academic calendars, leading to increased absenteeism and dropout rates
Energy	• Power shortages due to reduced hydropower generation in hydroelectric plants • Damage to energy infrastructure • Higher energy consumption, especially for cooling
Health	• Increased prevalence of malaria • Reduced water quality and sanitation leads to outbreak of waterborne diseases such as typhoid • Food insecurity contributes to higher rates of malnutrition, particularly among children, pregnant women, and individuals with compromised immune systems
Land and housing	• Damage to homesteads • Displacement of communities • Loss of lives • Land degradation through increased soil erosion
Manufacturing	• Reduced agricultural outputs affect the availability of raw materials for manufacturing processes • Damage to the manufacturing facilities and related infrastructure such as factories, warehouses, and transportation networks • Power shortages affect manufacturing operations and increases reliance on alternative, often more expensive energy sources
Social development	• Displacement of communities • Destruction of school and health facilities infrastructure • Loss of lives • Changing climate patterns contribute to the spread of diseases, such as malaria and waterborne illnesses, due to increased temperatures and flooding
Trade	• Reduced export of key commodities such as coffee, tea, and cocoa • Damage to infrastructure such as roads and bridges disrupts supply chains, increases transportation costs, and hampers the timely delivery of goods to both domestic and international markets • Exporters face challenges meeting stringent environmental and sustainability standards imposed by trading partners, potentially limiting market access for non-compliant products
Transportation	• Erosion and degradation of roads, bridges, and culverts • Flooded or damaged roads hinder the movement of goods and people, affecting trade and access to essential services • Transport disruptions increase the cost of goods and services due to delays and the need for alternative, often longer, routes • Damaged infrastructure and congestion lead to longer travel times and increased fuel consumption, contributing to higher greenhouse gas emissions
Water and environment	• Degradation of ecosystems such as wetlands, and forests • Habitat loss and a decline in biodiversity • Reduced availability of water resources • Deterioration of water quality • Soil degradation through erosion and nutrient depletion

4. VULNERABILITY AND RISK ASSESSMENT IN BUNAMWAYA DIVISION

Vulnerability to climate change refers to the degree to which a system, community, or individual is susceptible to, and unable to cope with, the adverse impacts of climate change, including climate variability and extremes (IPCC, 2014). Vulnerability is a function of exposure, the sensitivity of a household or livelihood to the stress associated with that exposure, and the adaptive capacity to recover from the impacts of that exposure.

*Vulnerability = (Exposure * Sensitivity) – Adaptive Capacity*

4.1 Exposure to climate hazards

Exposure refers to the extent to which a system is exposed to climate change-related hazards. Climate hazards refer to events or conditions resulting from climate or weather-related phenomena that have the potential to cause harm to people, property, and the environment (Zscheischler et al., 2020). Bunamwaya division is mostly exposed to drought, floods, hailstorms, lightning, and windstorms. These climate hazards are briefly described below.

4.1.1 Drought Hazard

Drought is defined as a recurrent feature of climate that occurs when there is an extended period of abnormal deficiency in precipitation (relative to what is considered normal) (Sivakumar & Wilhite, 2002). Drought in Bunamwaya division has been attributed to environmental degradation majorly cutting down of trees, and encroachment of wetland ecosystems. The impacts of drought include water crisis, reduced crop and livestock productivity, and increased food prices. The warmer temperatures also favour the survival of some pests and parasites thus an increase in pests, parasites, and diseases.

Drought in Bunamwaya division was assessed using the Standardized Precipitation Index (SPI). Rainfall data (1994-2024) from CHIRPS (Climate Hazards Group InfraRed Precipitation with Station data) was used to compute the SPI. The SPI was computed using the Drought Indices Calculator (DrinC). Drought was then categorized based on SPI values as described by (Guttman, 1999; McKee et al., 1993).

The biggest part of the division (58.1%) is exposed to low drought hazard. However, cells such as Busingiri Nyanama, Kisingiri, Nankinga, Lufuka, Ngobe, Nfufu, Kikumbi, Kirimanya, and Zana; are exposed to moderate drought hazard (**Figure 12**).

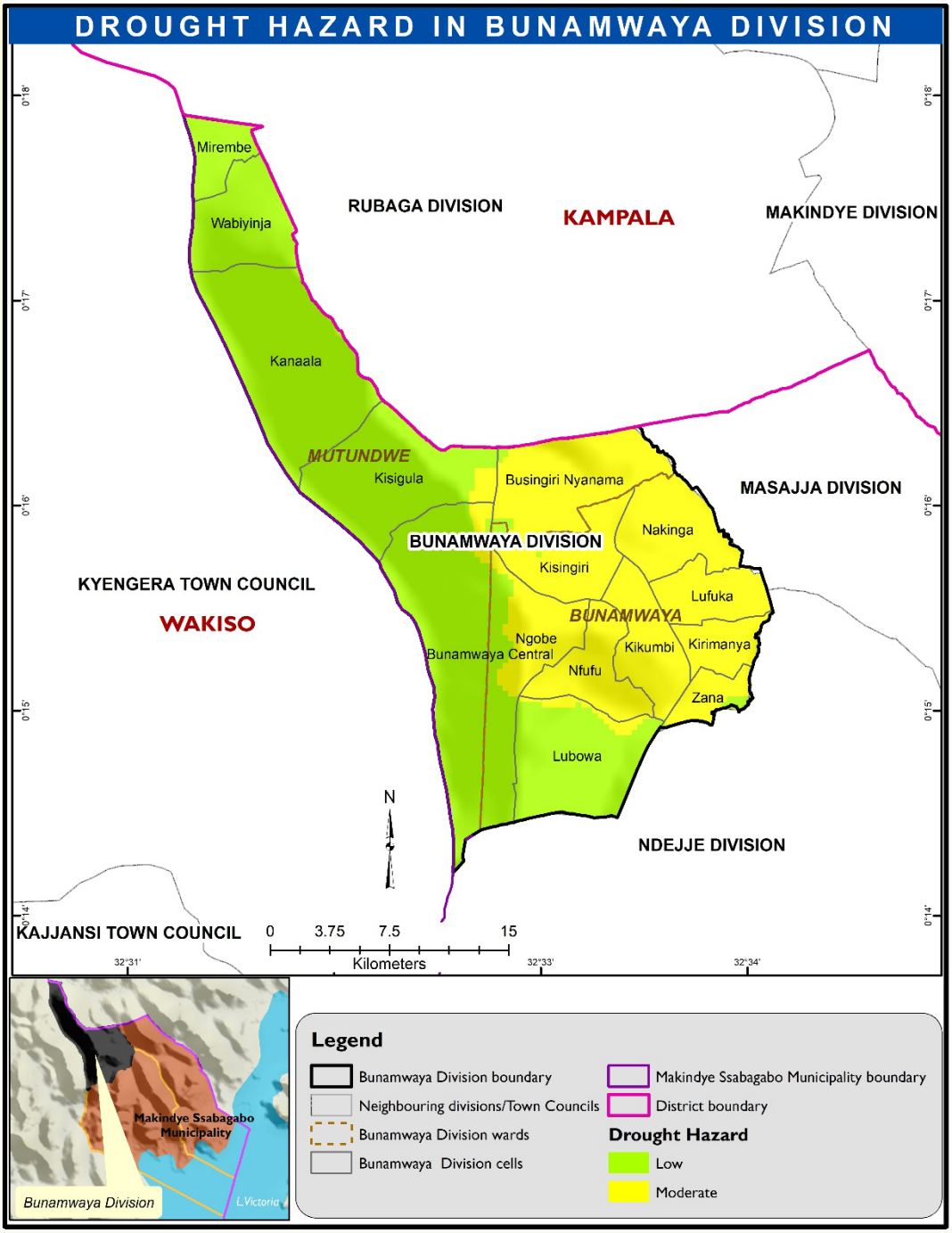


Figure 12: Drought hazard zonation in Bunamwaya division

4.1.2 Flood Hazard

Flooding is an overflow of water that submerges land that is usually dry (Farida & Maswanku, 2022). The floods in Bunamwaya division are majorly attributed to excessive rainfall due to climate change, clearing of trees and vegetation, encroachment of wetlands, municipal waste disposal in drainage channels, and increased farming of the river banks. The impacts of floods in Bunamwaya division include loss of lives; destruction of houses and household property; destruction of croplands; contamination of water sources; destruction of WASH facilities and infrastructure such

as health centres, schools, roads, bridges and culverts. The biggest part of the municipality (29.4%) is exposed to floods of very high intensity.

Flooding in Bunamwaya division was delineated using the HEC-GeoRAS tool. The input datasets included rainfall, elevation, land use/cover, and stream geometry data. The flood extents were then categorized into flood hazard intensities based on water depth: very high (>2m), high (1.5m-2m), moderate (1m-1.5m), and low (0.5m-1m). The flood modelling showed that due to wetland encroachment, most of the low-lying areas in Bunamwaya division are prone to floods. The areas that are prone to floods in Bunamwaya division include Mirembe, Wabiyinja, Kanaala, Kisigula, Bunamwaya Central, Lufuka, Kirimanya, Zana, and Nankinga (**Figure 13)**.

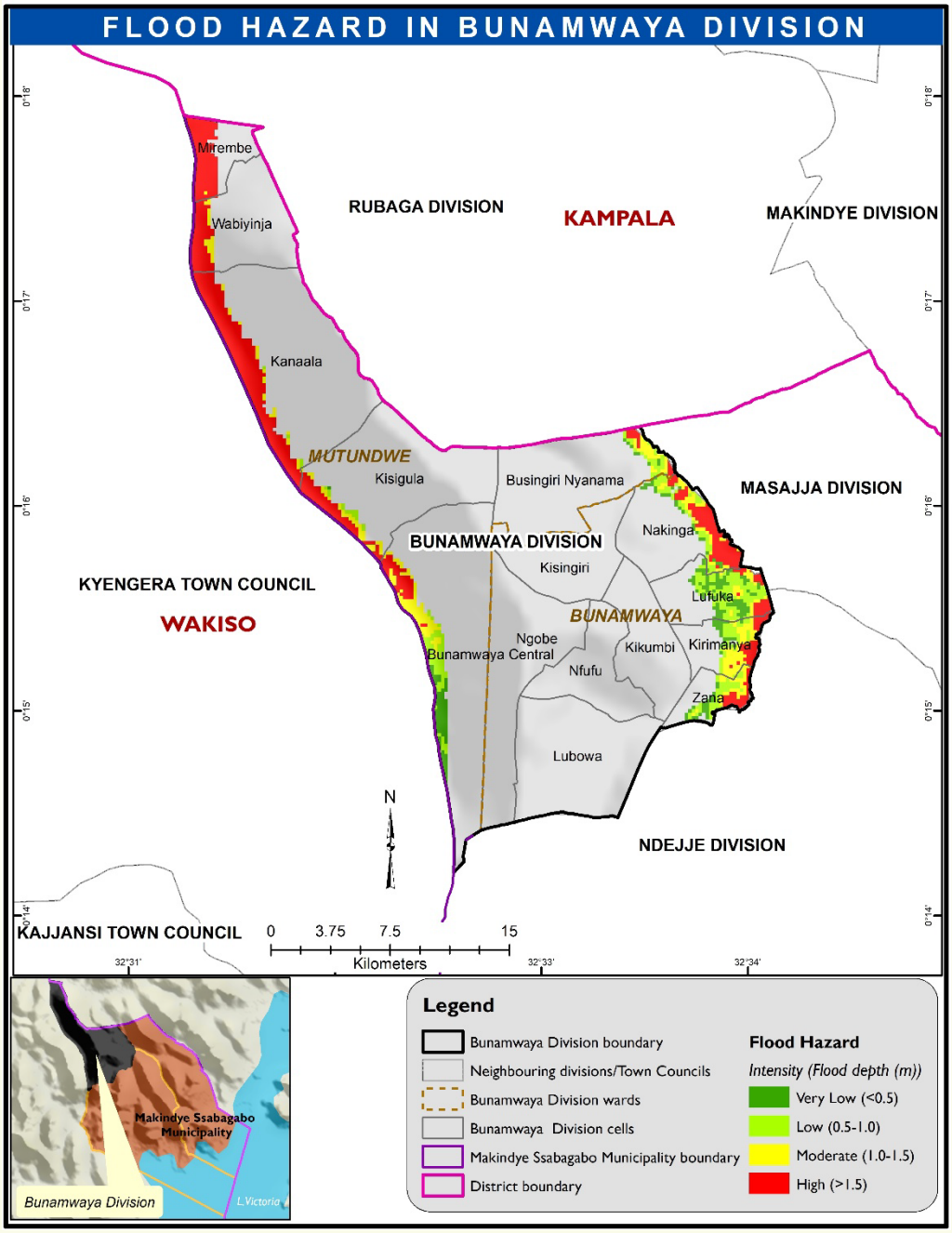


Figure 13: Flood hazard zonation in Bunamwaya division

4.1.3 Hailstorm Hazard

Hail is a form of solid precipitation. It consists of balls or irregular lumps of ice, each of which is called a hailstone. Hailstones consist mostly of water ice and measure between 0.2 inches and 6 inches in diameter. Hail develops when rising air in a thunderstorm, known as the updraft, lifts water droplets high into the atmosphere where temperatures are below freezing. This causes the water droplets to turn into hailstones before falling down to earth. The stronger the updraft, the larger the hailstones can become before succumbing to the pull of gravity and falling to the earth's surface. Depending on their intensity, hailstorms can destroy crops also cause temporary damage to WASH infrastructure (such as water points, and latrines). The entire division is moderately vulnerable to hailstorm hazard (**Figure 14**).

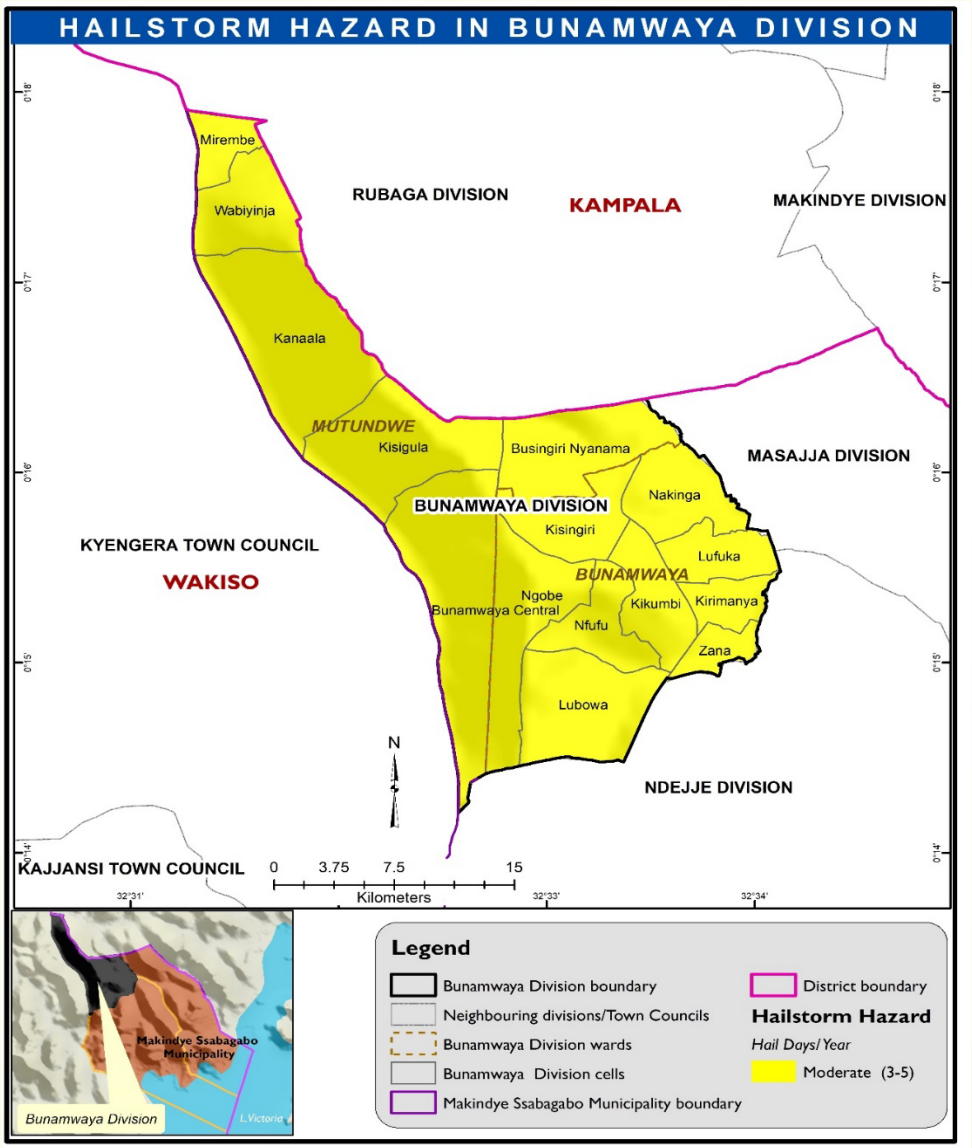


Figure 14: Hailstorm hazard zonation in Bunamwaya division
Data Source: OPM, 2019

4.1.4 Lightning Hazard

Lightning is the occurrence of a natural electrical discharge of very short duration and high voltage between a cloud and the ground or within a cloud (Agrawal & Nigam, 2014). The frequencies and severities of lightning incidences have increased, resulting in significant loss of life and property. Lightning happens under favourable conditions, where electrical discharges occur from a charge centre in a cloud either to: (i) the induced charge on the earth, (ii) the charge centres of another cloud, or (iii) a charge centre of the same cloud.

Thunder day data was used to profile lightning, and this was supplemented with Over-shooting Tops (OT) data acquired from NASA Meteosat Second Generation (MSG) 3km resolution Spin-Enhanced Visible Infrared Imager (SEVIRI). The lightning hazard zonation was generated with reference to flashes per square kilometre per year and classified as low, moderate, and high (**Table 2**). The biggest part of Bunamwaya division (55.7%) is exposed to low lightning hazard (**Figure 15**).

Table 2: Lightning hazard intensity scale

Intensity Class	OT Counts/month	Flash density (strikes/yr/km^2)
High	10-15	50-75
Moderate	5-10	10-50
Low	2-5	< 10

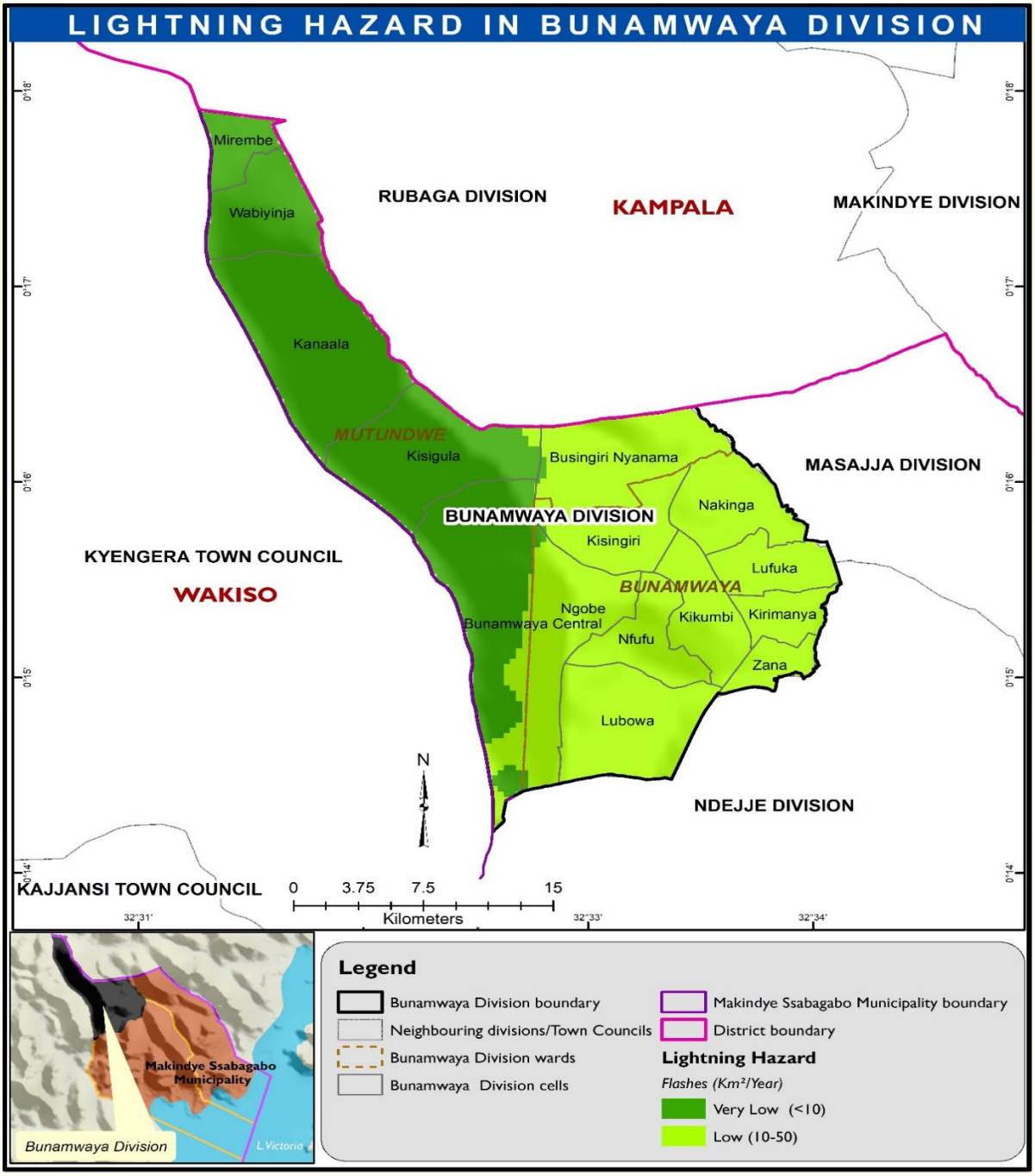


Figure 15: Lightning hazard zonation in Bunamwaya division

4.1.5 Windstorm Hazard

Wind is the perceptible natural movement of air. Wind is caused by differences in the atmospheric pressure. When a difference in atmospheric pressure exists, air moves from the higher to the lower pressure area, resulting in winds of various speeds. The impacts of windstorms in Bunamwaya division include air pollution, the spread of human diseases, uprooting of trees and considerable structural damage to buildings, power and telephone lines, communication masts and other urban infrastructure.

The assessment adopted the wind speed modelling approach by Morjani (2011) to develop the windstorm hazard of Bunamwaya division. Wind speed data for the municipality was obtained from the climate engine for the period 1994-2024. The spatial distribution of wind hazard intensity was done by conducting a kriging interpolation analysis and applying the Beaufort wind scale (**Table 3**).

Table 3: Beaufort wind hazard intensity scale

Beaufort scale	Description	Wind speed (m/s)	Conditions	Windstorm Hazard Class
6	Strong breeze	10.8-14.5	Large tree branches/crops in motion; whistling heard in electric/telegraph wires; difficulty using umbrellas	Very low
7	High wind, moderate gale	14.6-20.0	Whole trees in motion; difficulty when walking against the wind	Low
8	Gale	20.1-22.0	Twigs break off trees; generally, impedes progress	Moderate
9	Strong/severe gale	22.1-28.0	Can cause slight structural damages	High
10	Storm, whole gale	> 28.1	Trees/crops uprooted; considerable structural damage, may be accompanied by widespread damage	Very high

The biggest part of the division (96.1%) is exposed to very low windstorm hazard (**Figure 16**).

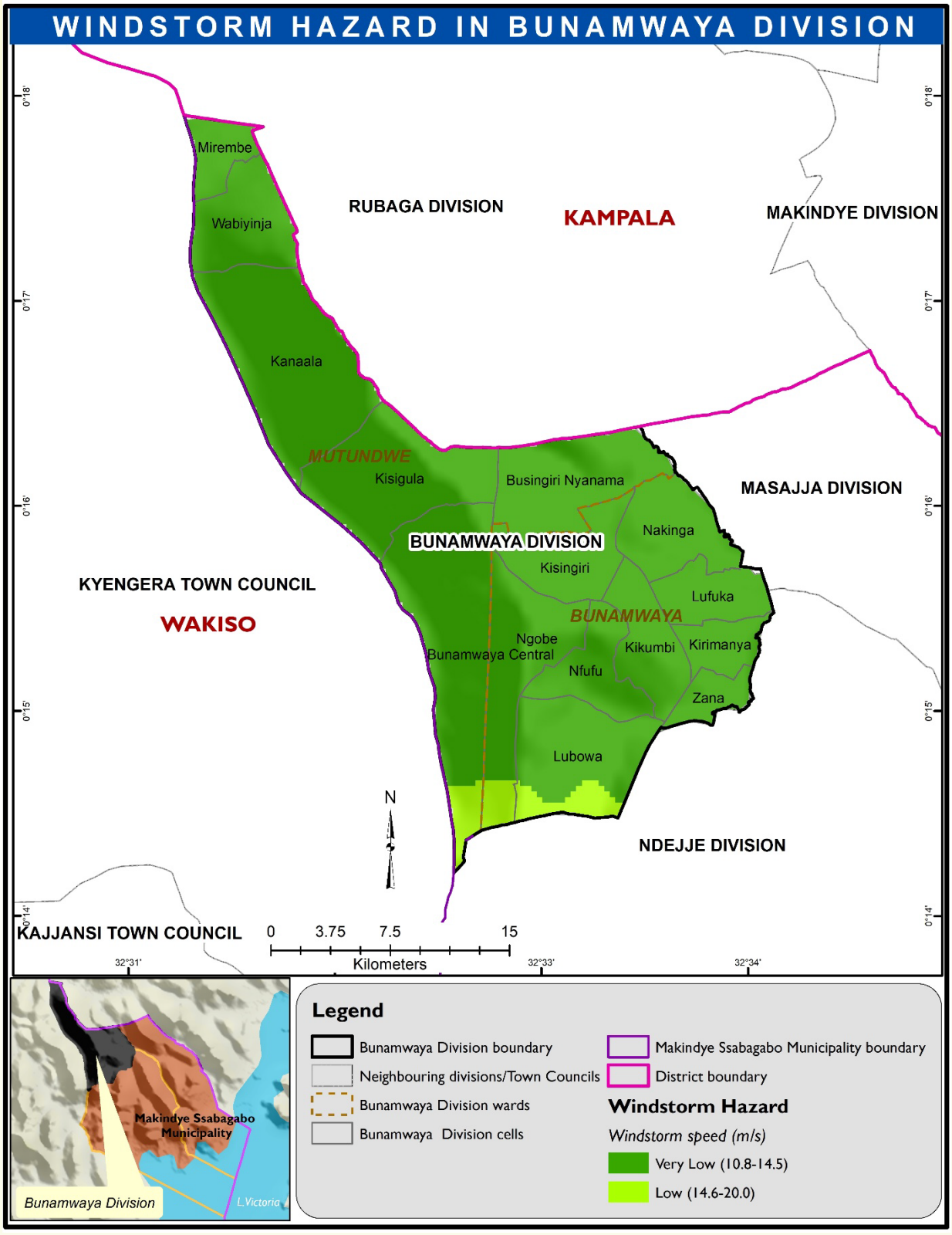


Figure 16: Windstorm hazard zonation in Bunamwaya division

4.1.6 Multi-hazard

Multi-hazard profiling considers more than one hazard in a given place and the interrelations between these hazards, including their simultaneous or cumulative occurrence and their potential

interactions. The assessment showed that the areas that have a high level of exposure (more than 80 percent) to all the climate hazards in Bunamwaya division include Mirembe, Wabiyinja, Kanaala, Busingiri Nyanama, Nakinga, Lufuka, Kirimanya, and Zana (**Figure 17**).

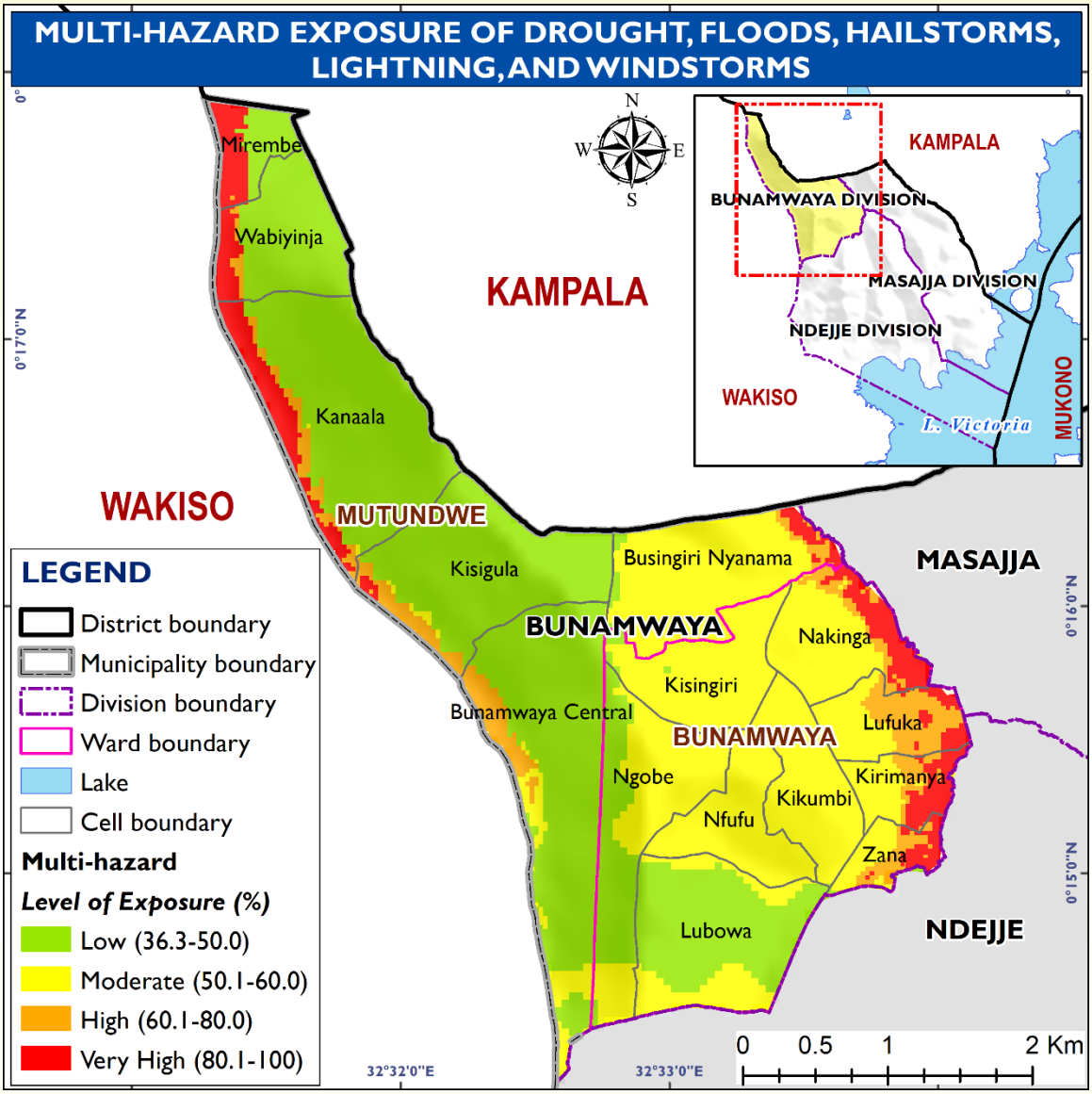


Figure 17: Multi-hazard exposure to drought, floods, lightning, hailstorms, and windstorms in Bunamwaya division

4.2 Sensitivity of communities to multi-hazards

Sensitivity describes the socio-economic or demographic characteristics that can make people susceptible to the negative effects of exposure (Declet-Barreto et al., 2020). The sensitivity indicators that were assessed at cell level are presented in **Table 4**. These indicators were ranked, weighed and normalized to produce the sensitivity index.

Table 4: Sensitivity indicators to multi-hazards in Bunamwaya division

No	Sensitivity indicators	Description	Data source	Hotspot cells
1.	Access to health services	Access to health services reflects a community's ability to prevent, respond to, and recover from climate-related health impacts such as heat stress, disease outbreaks, and injuries from extreme weather.	Key Informant Interviews, 2025	Zana, Ngobe, Nfufu, Kanaala, and Kisigula
2.	Community awareness and preparedness	Community awareness and preparedness reflects how well individuals and groups understand climate risks and their capacity to take timely action.	Key Informant Interviews, 2025	Moderate in the entire division
3.	Degradation of rivers	Degradation of rivers signifies the reduced ability of freshwater ecosystems to absorb and adapt to climate impacts, thereby increasing the vulnerability of communities that rely on these rivers for water, food, and livelihoods.	Key Informant Interviews, 2025	Lubowa, Kirimanya, Nakinga, Lufuka, Wabiyinja, Bunamwaya Central, Mirembe, Busingiri Nyanama, Zana, Kanaala, and Kisigula
4.	Dependency on climate-sensitive sectors e.g., agriculture	Dependency on climate-sensitive sectors highlights how heavily communities rely on industries like agriculture, fisheries, and tourism, which are directly affected by climate variability, making them more vulnerable to disruptions caused by extreme weather events, changing temperatures, and shifting ecosystems.	Key Informant Interviews, 2025	Bunamwaya Central, and Ngobe
5.	Dependence on surface water	Dependency on surface water reveals how reliant communities are on rivers, wetlands, and reservoirs that are highly vulnerable to changing climate, increasing their risk of water scarcity and reduced water quality under climate stress.	Key Informant Interviews, 2025	Low in the entire division
6.	Drainage systems	The design and capacity of drainage systems determine how well a community can manage increased rainfall, flooding, and storm surges, with inadequate or poorly maintained systems heightening vulnerability to climate-induced water-related hazards.	Key Informant Interviews, 2025	Poor in the entire division
7.	Housing quality	Housing quality reflects how well homes can protect occupants from climate-related hazards like floods, with poor-quality housing increasing the risk of damage, displacement, and health impacts during climate shocks.	Key Informant Interviews, 2025	Moderate in the entire division
8.	Income levels	Income levels influence a community's ability to prepare for, respond to, and recover from climate impacts, with lower-income populations often lacking the resources needed for adaptation,	UBOS, 2021	Moderate in the entire division

No	Sensitivity indicators	Description	Data source	Hotspot cells
		making them more vulnerable to climate-related risks.		
9.	Population density	High-density areas can amplify the impacts of climate hazards such as heatwaves, and disease spread while straining infrastructure and emergency response systems, increasing overall community vulnerability.	UBOS, 2024	Kirimanya, Lubowa, Lufuka, and Zana
10.	Unemployment	Unemployment reflects limited economic resilience and reduced access to resources, making individuals and communities less able to adapt to or recover from climate-related events such as floods	Key Informant Interviews, 2025	Moderate in the entire division
11.	Wetland degradation	Wetland degradation weakens the natural buffers that wetlands provide against climate impacts like flooding, storm surges, and droughts, reducing ecosystem resilience and increasing the vulnerability of both human and ecological communities that depend on them.	WMD, 2021	Kirimanya, Kisingiri, Lufuka, Zana, Kanaala, Nakinga, Wabiyinja, Busingiri Nyanama, and Bunamwaya Central

The cells that have a high sensitivity to multi-hazards (i.e., drought, floods, hailstorms, lightning, and windstorms) in Bunamwaya division include Mirembe, Kanaala, Bunamwaya Central, and Kirimanya (**Figure 18**).

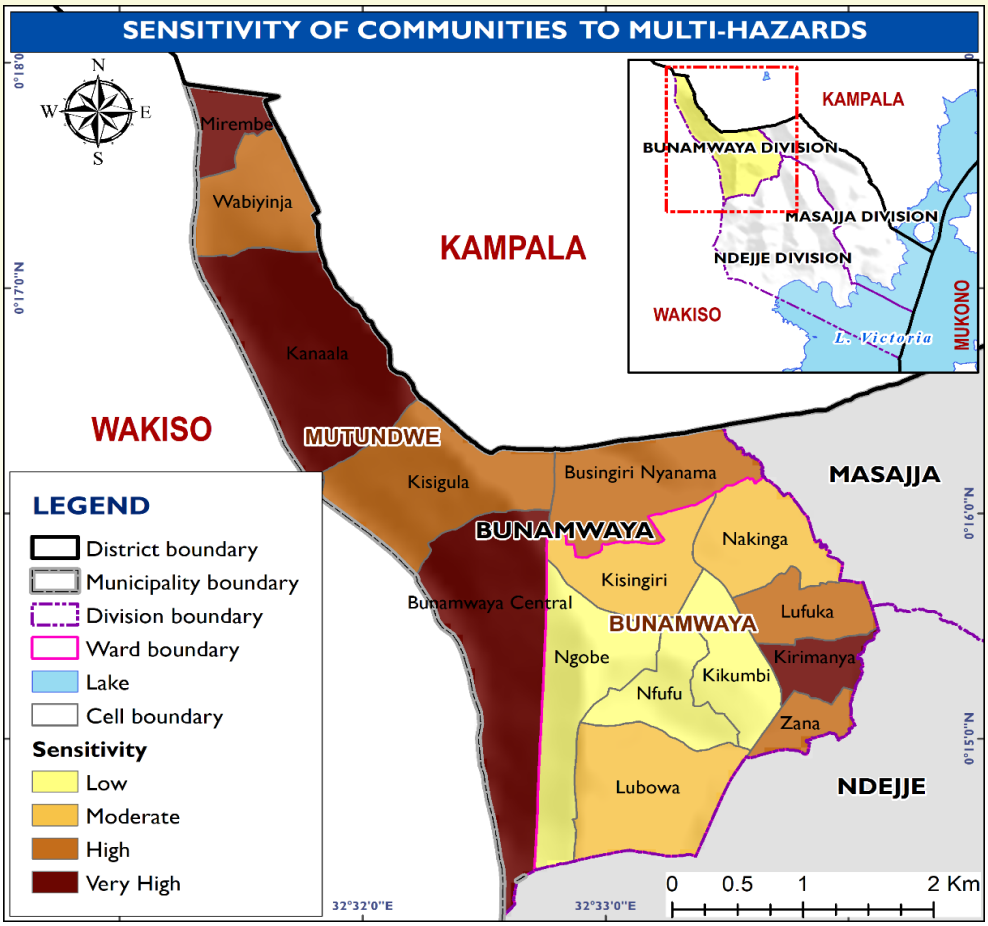


Figure 18: Sensitivity of communities to multi-hazards in Bunamwaya division

4.3 Adaptive capacity of communities to multi-hazards

Adaptive capacity is defined as the ability of systems, institutions, humans, and other organisms to adjust to potential damage, to take advantage of opportunities, or to respond to consequences of climate change (IPCC, 2023). The adaptive capacity indicators that were assessed at the cell level are shown in **Table 5**. These adaptive capacity indicators were ranked, weighed and normalized to produce the adaptive capacity index.

Table 5: Adaptive capacity indicators to multi-hazards in Bunamwaya division

No	Adaptive capacity indicators	Description	Data source	Hotspot cells
1.	Access to clean water and sanitation	Access to clean water and sanitation ensures that communities can maintain good health and hygiene, reducing the risk of waterborne diseases during extreme weather events, and enabling better resilience to climate-induced health challenges.	Key Informant Interviews, 2025	High in the entire division
2.	Access to credit	Access to credit allows individuals and communities to invest in resilience-building measures, such as climate-smart agriculture, infrastructure improvements, or emergency preparedness, providing the financial	Key Informant Interviews, 2025	Moderate in the entire division

No	Adaptive capacity indicators	Description	Data source	Hotspot cells
		resources needed to recover and adapt to climate-related disruptions.		
3.	Access to early warning information	Access to early warning information enables communities to prepare for and respond effectively to climate-related hazards, such as storms, floods, or heatwaves, by providing timely alerts and guidance that help reduce risks, protect lives, and minimize damage.	Key Informant Interviews, 2025	Low in the entire division
4.	Access to renewable energy e.g., solar	Access to renewable energy provides communities with a sustainable and reliable energy source, reducing dependence on fossil fuels, minimizing greenhouse gas emissions, and enhancing resilience by ensuring energy availability during climate-related disruptions like storms or power outages.	Key Informant Interviews, 2025	Kisingiri, Ngobe, Nfufu, Kikumbi, Kirimanya, Nakinga, Lufuka, Kisigula, and Mirembe
5.	Awareness of climate risks and adaptation practices	Awareness of climate risks and adaptation practices reflects how well individuals and communities understand climate-related challenges and the actions they can take to mitigate risks, enabling them to make informed decisions and implement strategies that enhance resilience to changing climate conditions.	Key Informant Interviews, 2025	Moderate in the entire division
6.	Diverse livelihood options	Diverse livelihood options provide communities with multiple sources of income and resources, allowing them to better withstand climate shocks, and adapt by shifting to more resilient economic activities when needed.	Key Informant Interviews, 2025	Moderate in the entire division
7.	Education levels	Higher levels of education enable individuals and communities to better understand climate risks, access information, and adopt effective adaptation strategies, thereby enhancing their ability to respond to and recover from climate-related challenges.	Key Informant Interviews, 2025	High in the entire division
8.	Extension services	Extension services provide farmers, communities, and local businesses with essential knowledge, skills, and resources to adopt climate-resilient practices, improve productivity, and enhance the ability to cope with changing climate conditions.	Key Informant Interviews, 2025	High in the entire division
9.	Good environmental management practices	Good environmental management practices promote sustainable resource use, reduce environmental degradation, and enhance ecosystem resilience, enabling communities to better withstand and adapt to climate impacts.	Key Informant Interviews, 2025	Moderate in the entire division

No	Adaptive capacity indicators	Description	Data source	Hotspot cells
10.	Healthcare infrastructure	Robust and well-maintained healthcare systems improve a community's ability to respond to climate-related health challenges, such as heatwaves, disease outbreaks, and extreme weather events, by providing timely medical care, disease prevention, and emergency services.	Key Informant Interviews, 2025	Kisingiri, Ngobe, Nfufu, Kikumbi, Kirimanya, Nakinga, and Lufuka
11.	Proper waste disposal	Proper waste disposal helps prevent environmental degradation, reduces the risk of disease outbreaks, and ensures cleaner, healthier living conditions, enabling communities to better withstand the health and environmental impacts of climate change, such as flooding	Key Informant Interviews, 2025	Kirimanya, Lufuka, Kisingiri, Ngobe, Nfufu, Kikumbi, and Nakinga
12.	Rainwater harvesting	Rainwater harvesting provides an alternative and supplemental water source, reducing dependence on traditional sources such as rivers especially during the dry season.	Key Informant Interviews, 2025	Kirimanya, Lufuka, Kisingiri, Ngobe, Nfufu, Kikumbi, Nakinga, Kisigula, and Mirembe
13.	Social networks	Strong, connected communities are better able to share resources, information, and support during climate events, helping individuals and groups recover more quickly and effectively from climate impacts	Key Informant Interviews, 2025	Kisigula, Mirembe, Zana, Lubowa, Kanaala, Wabiyinja, Bunamwaya Central, and Busingiri Nyanama
14.	Urban greening initiatives	Urban greening initiatives enhance cities' resilience by increasing green spaces, improving air quality, reducing heat island effects, and providing natural buffers against extreme weather, all of which help communities better cope with climate-related challenges.	Key Informant Interviews, 2025	Moderate in the entire division
15.	Wetland and ecosystem restoration	Wetland and ecosystem restoration enhances the natural ability of wetlands to absorb climate impacts such as flooding, while supporting biodiversity, improving water quality, and strengthening the resilience of communities that rely on these ecosystems.	Key Informant Interviews, 2025	Low in the entire division

Zana, Lubowa, Wabiyinja, Kanaala, Busingiri Nyanama, and Bunamwaya Central have a very high adaptive capacity to multi-hazards (i.e., drought, floods, hailstorms, lightning, and windstorms) while Lufuka and Kirimanya have a low adaptive capacity to multi-hazards (**Figure 24**).

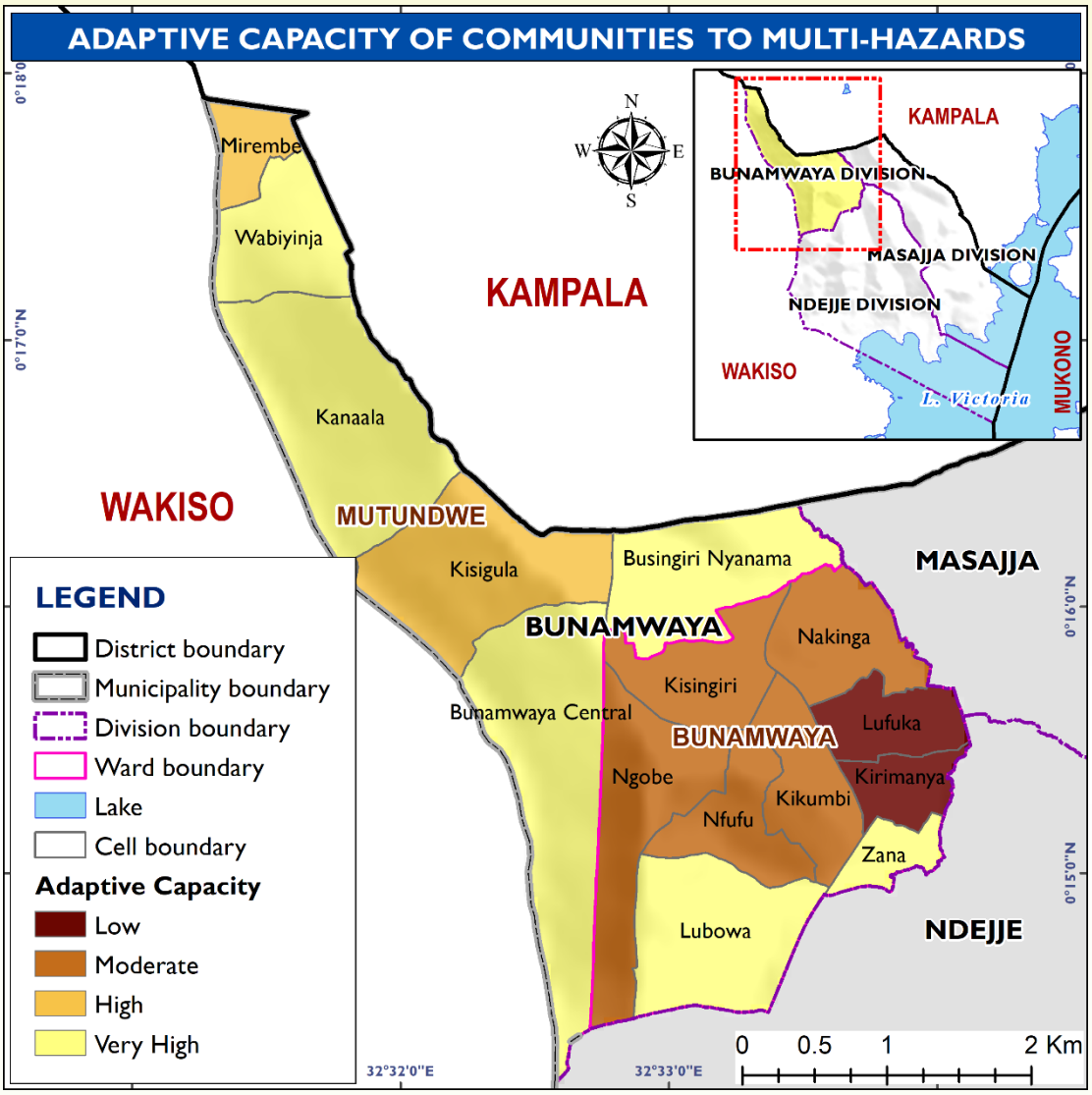


Figure 19: Adaptive capacity of communities to multi-hazards in Bunamwaya division

4.4 Vulnerability of communities to multi-hazards

The communities that are highly vulnerable to multi-hazards (i.e., drought, floods, hailstorms, lightning, and windstorms) in Bunamwaya division are found in Mirembe, Nakinga, Lufuka, Kirimanya, and Zana (**Figure 20**).

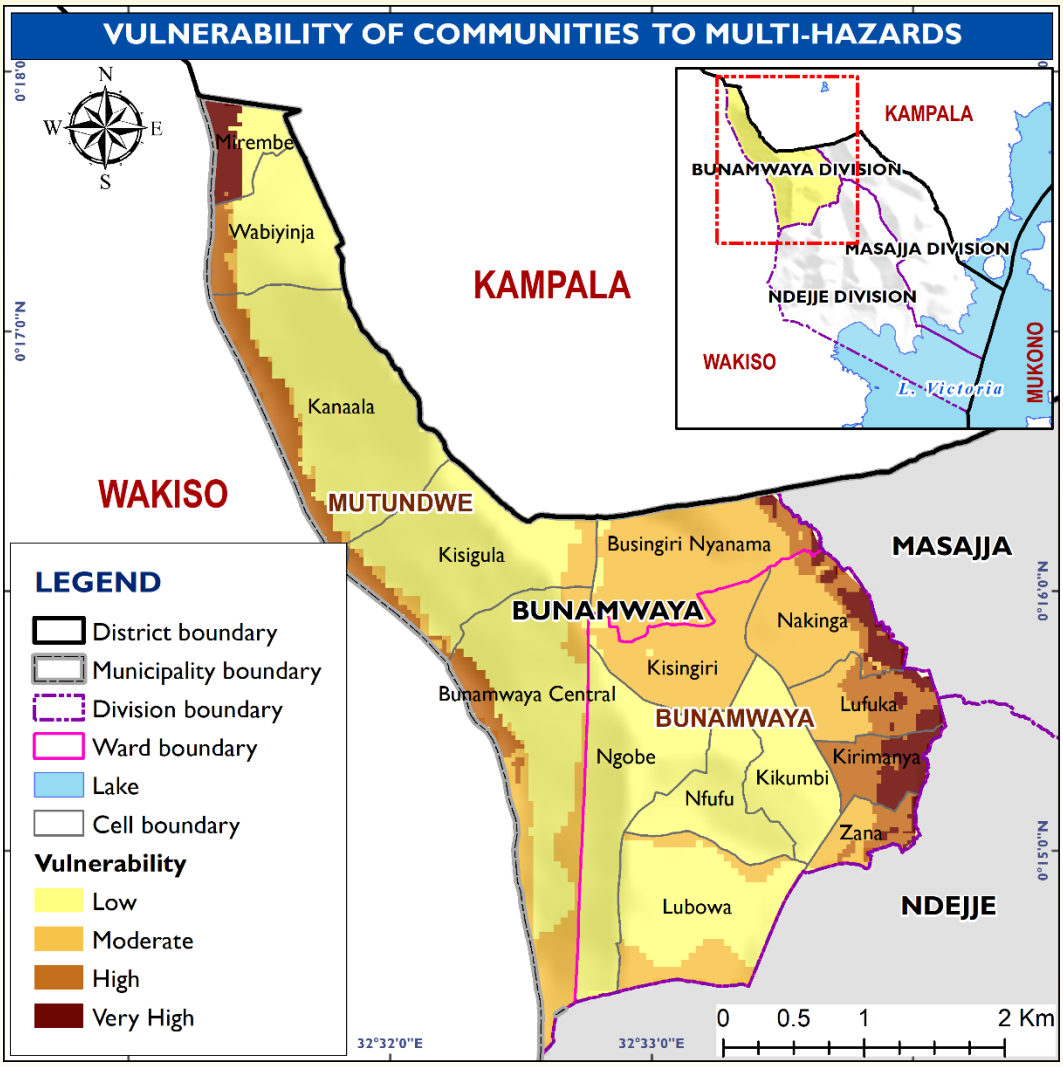


Figure 20: Vulnerability of communities to multi-hazards in Bunamwaya division

4.5 Risk of communities to multi-hazards

Disaster risk is expressed as the likelihood of loss of life, injury or destruction and damage from a disaster in a given period (UNISDR, 2017). Disaster risk is widely recognized as the consequence of the interaction between a hazard and the characteristics that make people and places vulnerable and exposed (**Figure 21**).



Figure 21:The risk matrix
Source: de Brito et al., 2017

The multi-hazards experienced in Bunamwaya division were combined, weighed, and normalized to produce the hazard index. The population of the communities in the different cells was considered as the exposed element. The population was normalized to produce the exposure index. The hazard, exposure, and vulnerability indices were analyzed and normalized to produce the risk index, which was mapped at the cell level.

Risk assessment showed that the cells that are at a high risk of multi-hazards (drought, floods, hailstorms, lightning, and windstorms) in Bunamwaya division include Mirembe, Wabiyinja, Kanaala, Kisigula, Nakinga, Busingiri Nyanama, Lufuka, and Kirimanya (**Figure 22**).

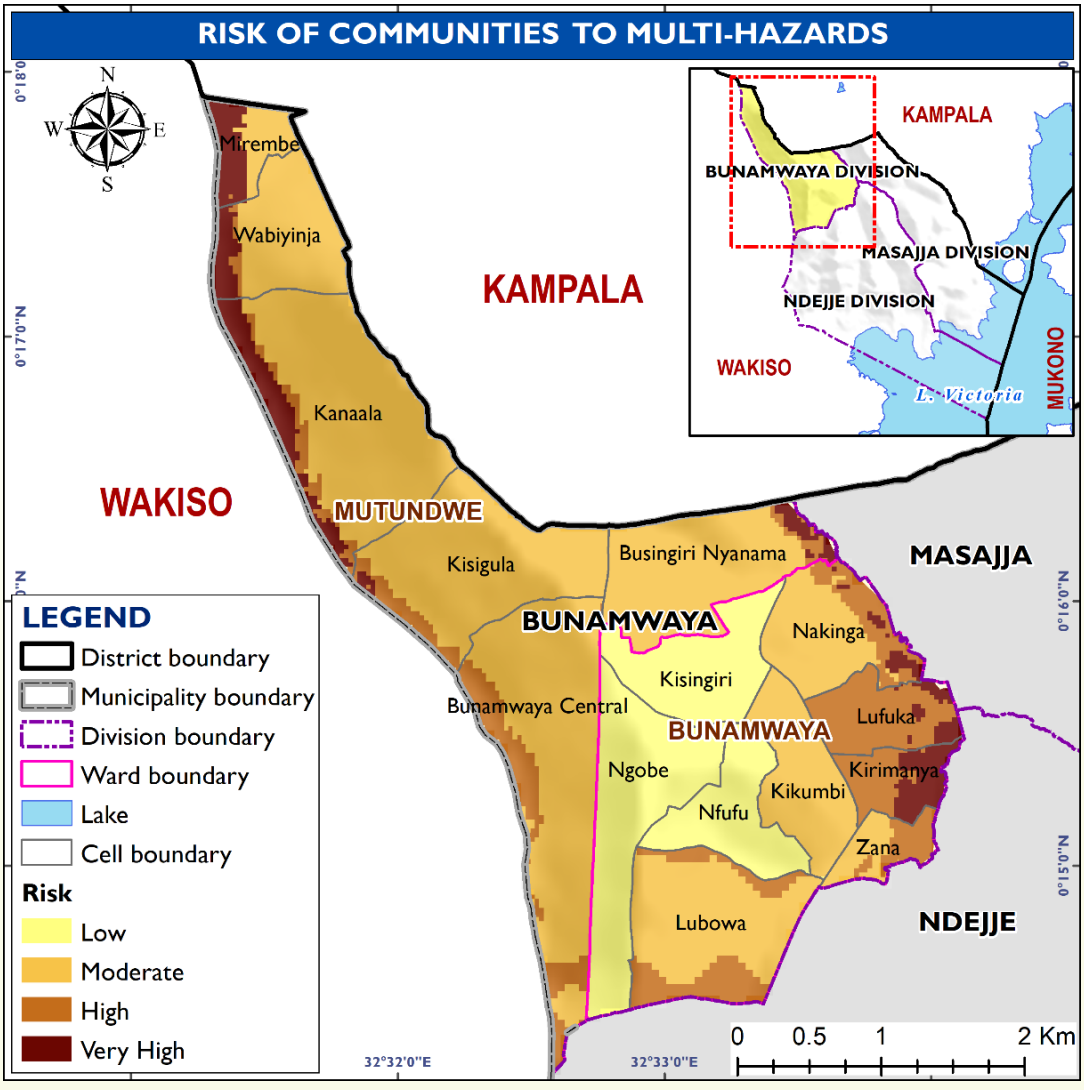


Figure 22: Risk of communities to multi-hazards in Bunamwaya division

5. RISK REDUCTION MEASURES TO CLIMATE HAZARDS

5.1 Introduction

This chapter provides disaster risk reduction measures that are implemented by communities in Bunamwaya division to cope or mitigate the negative impacts of disasters. The measures are aimed at improving the resilience of communities to disaster impacts and enhance their livelihoods.

Disaster Risk Reduction (DRR) measures are referred to as a combination of activities that mitigate the adverse impacts of natural hazards and increases community resilience by:

I. Identifying and analyzing hazards and threats: Supporting vulnerable communities to understand major hazards they face.
II. Reducing exposure and vulnerabilities: Addressing the conditions that make communities susceptible to those hazards and shocks they have identified.
III. Controlling and treating risks: helping communities address the potential damage and losses caused by the hazards and threats and preventing hazardous events from becoming disasters to the communities.
IV. Enhancing capacities: building capacity of individuals, communities and institutions to reduce risk and build greater resilience.

DRR measures can be broadly classified as those that are policy and planning related (non-structural) and physical measures (structural).

I. Policy and planning measures are implemented at the national and sub-national levels and help to integrate DRR into the policy and institutional frameworks.
II. Physical measures are designed to reduce the exposure and vulnerability of infrastructure to natural hazards (prevention) as well as to provide coping and adaptive infrastructure in case of a disaster event (coping/adaptive).

5.2 Hazard-specific Disaster Risk Reduction Measures

The structural and non-structural disaster risk reduction measures for each climate hazard in Bunamwaya division are highlighted in **Table 6** below.

Table 6: Structural and non-structural DRR measures in Bunamwaya division

No	Hazard	Structural Measures	Non-Structural Measures
1	Drought	1. Plant drought-resistant and quick-maturing crop varieties. 2. Implement small-scale irrigation systems. 3. Demarcate wetlands to reduce encroachment. 4. Promote urban greening (tree planting, green corridors) to reduce heat and erosion. 5. Adopt alternative sources of livelihood such as cultural tourism, crafts and apiary 6. Construct community water reservoirs and underground tanks.	1. Sensitize communities about water-saving technologies, water and soil conservation practices, and environmental degradation. 2. Implement good environmental management practices 3. Encourage use of drought-tolerant crops 4. Promote rainwater harvesting systems on schools, health centers, and households 5. Encourage farmers to purchase crop insurance.

No	Hazard	Structural Measures	Non-Structural Measures
2	Floods	1. Relocate vulnerable communities to safer areas. 2. Reforest and replant native vegetation in degraded wetland areas. 3. Limit or restrict development in floodplains. 4. Construct large drainage channels especially in flood-prone areas. 5. Desilt and expand existing drainage channels. 6. Promote raised foundations and water-resistant building materials 7. Develop, design and implement urban settlement plans. 8. Implement community-led solid waste management to prevent drain blockages	1. Develop and implement an ordinance on River bank management to regulate human activities along river banks. 2. Sensitize communities about flood risk and safety as well as sustainable land management. 3. Develop and implement catchment and community-based wetland management plans. 4. Incorporate flood mitigation in local planning. 5. Protect and restore flood natural mitigation features such as wetlands, river banks and hill/mountain slopes 6. Use indigenous knowledge in planning and management of floods. 7. Use low-cost sensors and community reporting to monitor water levels. 8. Restoration of watersheds through FMNR.
3	Windstorms	1. Plant windbreaks (trees and shrubs) around vulnerable settlements and open areas. 2. Implement climate smart agriculture practices like agroforestry. 3. Provide technical guidance for retrofitting existing structures.	1. Increase awareness about severe windstorms. 2. Support enforcement of building codes to strengthen roof anchoring and wall bracing 3. Install an effective monitoring and early warning system about windstorms. 4. Enforce use of safety gears like life jackets and use motorised boats
4	Hailstorms	1. Use strong roofing materials 2. Maintain good drainage for road networks 3. Plan early to escape the hailstorms. 4. Create a local contingency fund for agricultural recovery	1. Install an effective monitoring and early warning system about hailstorms. 2. Avoid walking out of the house during hailstorms. 3. Increase hail risk awareness 4. Encourage mixed cropping and agroforestry to reduce impact
5	Lightning	1. Install lightning arresters on critical buildings. 2. Enhance building codes. 3. Plant trees	1. Conduct lightning awareness programs and education (e.g., avoiding shelter under trees). 2. Install an effective monitoring and early warning system about lightning. 3. The government should subsidize lightning arrestors.

6. CONCLUSION AND RECOMMENDATIONS

6.1 Conclusion

Over time, Bunamwaya division has experienced climate hazards. These include drought, floods, windstorms, hailstorms, and lightning. However, the frequency and severity of these hazards has increased in the past years due to changes in climate. This has increased the vulnerability and risk of elements and communities to multi-hazards. The cells that are highly vulnerable and at risk of multi-hazards in Bunamwaya division include Mirembe, Nakinga, Lufuka, Kirimanya, Zana, Wabiyinja, Kanaala, Kisigula, and Busingiri Nyanama. Therefore, the Government with support from partners should design innovative approaches to reduce the vulnerability and risk of these communities to multi-hazards and also increase their resilience and livelihoods.

6.2 Recommendations

The recommendations are categorized into four themes and these include understanding disaster risk; improving disaster risk governance; investing in disaster risk reduction, preparedness, and early warning; and building back better in the aftermath of disaster.

Understanding disaster risk

- Create a centralized GIS-based platform for mapping local climate hazards
- Conduct participatory risk mapping and engage communities to validate and update local hazard data using indigenous knowledge.
- Regularly monitor and update hazard data e.g., through routine post-disaster data reviews and assessments
- Use schools, radio, and local leaders to educate residents about climate-related risks and safety practices.

Improving disaster risk governance

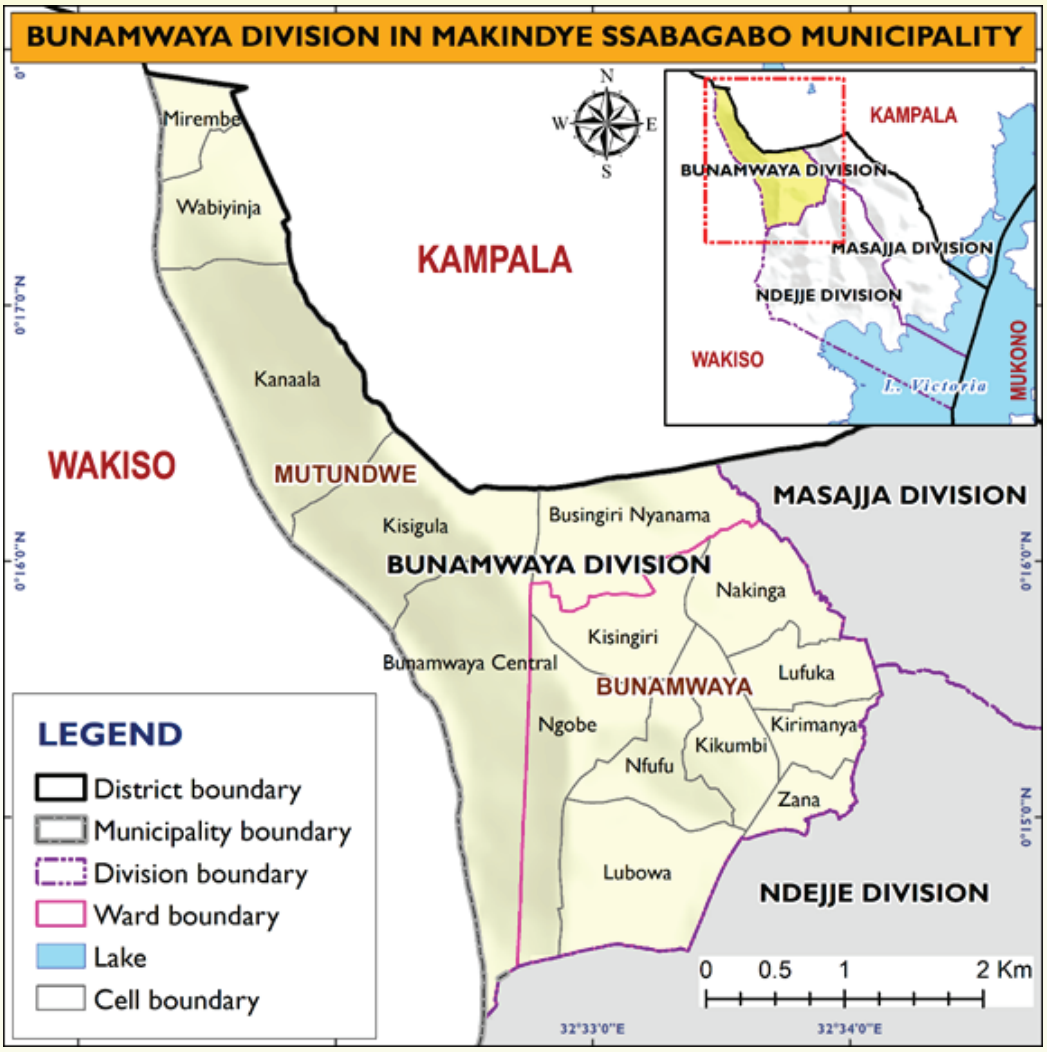
- Use hazard-specific information to guide land use, zoning, and infrastructure investments.
- Train local government staff (e.g., planners, engineers, and councilors) in climate risk analysis and resilient urban planning.
- Establish a local disaster risk coordination committee that facilitates cross-sectoral coordination and accountability for risk reduction efforts.
- Align budgets with risk priorities. This can include allocating local funds based on identified vulnerability hotspots and community needs.

Investing in disaster risk reduction, preparedness, and early warning

- Upgrade and regularly maintain flood-prone drainage systems and culverts.
- Use ecosystem-based adaptation to mitigate flood impacts and recharge water sources.
- Install lightning arrestors on public facilities such as schools, health centers, and markets.
- Promote water harvesting structures (such as rooftop tanks, underground reservoirs, and community water banks) for drought resilience.
- Develop effective localized flood and storm alert systems via SMS, radio, and community leaders.
- Enforce building codes for wind- and flood-resilient construction and retrofitting.
- Equip local response units with tarpaulins, tools, and essential items for rapid deployment.

Building back better in the aftermath of disasters

- Provide fast-track access to seeds and tools after hailstorm or drought events.
- Promote stronger building materials, elevated housing, and storm-resistant roofing.
- Train local teams to evaluate damage, estimate needs, and coordinate response.
- Ensure that rebuilding also addresses social protection such as water access, secure housing, and livelihood support.



for more information please contact us
Makindye Ssabagabo Municipal Council Resource Pool
P.O.Box 1872, Kampala

GREATER KAMPALA METROPOLITAN AREA URBAN DEVELOPMENT PROGRAM

THE WORLD BANK